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Questions

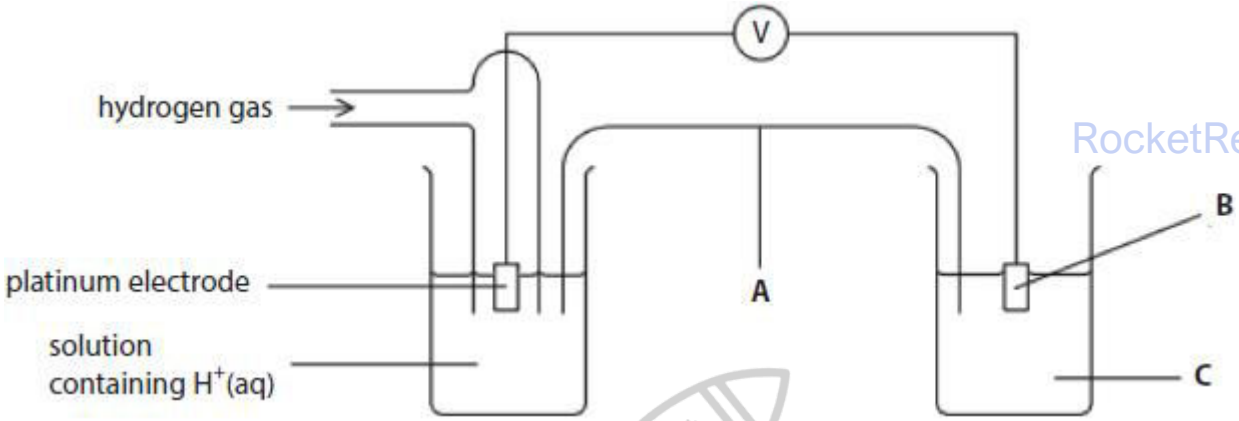
Q1.

This question is about electrochemical cells.

(a) The half-equation for an iron(III) / iron(II) half-cell is

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$$

The standard electrode potential of this half-cell is measured using the apparatus shown.



The measurement is made under standard conditions.
Identify, by name or formula, the substances needed in the salt bridge and in the right-hand half-cell to measure the standard electrode potential.

(3)

A Salt bridge containing a solution of

.....

B Electrode made of

.....

C Solution containing

.....

(b) The cell diagram for an electrochemical cell is

$$Pt \mid [Mn^{2+} + 4H_2O], [MnO_4^- + 8H^+] \,\|\, [BiO_3^- + 6H^+], [Bi^{3+} + 3H_2O] \mid Pt \quad E^{\ominus}_{cell} = +0.09\ V$$

Write the half-equations for the two half-cells and hence the overall ionic equation for the reaction that occurs.
State symbols are not required.

(3)

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(c) The standard electrode potential, $E^{\ominus}$, for the chromium(III) / chromium half-cell is −0.74 V.

$$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s)$$

The effect on the electrode potential, *E*, of changing the concentration of the ions in the half cell is calculated using the equation

$$E = E^{\ominus} + \frac{RT}{96\,500 \times n} \ln[Cr^{3+}(aq)]$$

where n is the number of electrons in the half-equation, *T* is the temperature in kelvin and *R* is the gas constant.
Calculate the electrode potential of the half-cell at 298 K when the concentration of chromium(III) ions is 0.0100 mol dm^{-3}.

(2)

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(Total for question = 8 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many isomers containing a benzene ring are there with the molecular formula $C_6H_3(NO_2)_2OH$?

- ☐ **A** 4
- ☐ **B** 5
- ☐ **C** 6
- ☐ **D** 7

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(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In acidic conditions, manganate(VII) ions are reduced to manganese(II) ions, Mn^{2+}.

In neutral conditions, manganate(VII) ions are reduced to manganese(IV) oxide, MnO_2.

22.0 cm^3 of a solution of manganate(VII) ions were needed to oxidise 25.0 cm^3 of a solution of iron(II) ions to iron(III) ions in **acidic** conditions.

The same solution of manganate(VII) ions is used to oxidise 25.0 cm^3 of the same solution of iron(II) ions in **neutral** conditions.

By considering the oxidation number changes involved, it may be shown that the volume of the manganate(VII) solution required, in neutral conditions, is

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- ☐ **A** 12.6 cm^3
- ☐ **B** 13.2 cm^3
- ☐ **C** 36.7 cm^3
- ☐ **D** 38.5 cm^3

(Total for question = 1 mark)

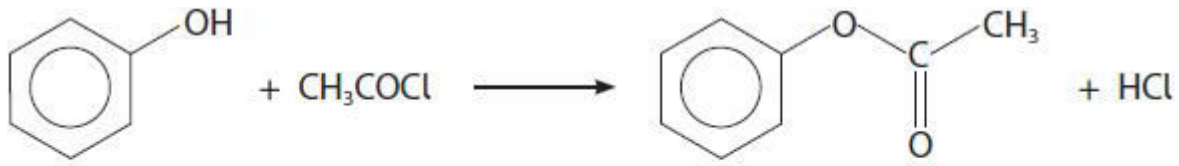
Q4.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Phenol reacts with ethanoyl chloride to form phenyl ethanoate.



phenol

$M_r = 94$

phenyl ethanoate

$M_r = 136$

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In an experiment, 3.67 g of phenyl ethanoate was formed, giving a yield of 85.0 %.

What was the starting mass of phenol?

- ☒ **A** 2.16 g
- ☒ **B** 2.54 g
- ☒ **C** 2.98 g
- ☒ **D** 4.51 g

Use this space for any rough working. Anything you write in this space will gain no credit.

(Total for question = 1 mark)

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Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A hydrocarbon contains 91.3 % by mass of carbon.

Which of these could be the molecular formula of this compound?

- ☒ **A** C_6H_6
- ☒ **B** C_7H_8
- ☒ **C** C_8H_{10}
- ☒ **D** C_9H_{12}

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(Total for question = 1 mark)

Q6.

This question is about some organic compounds.

(a) A mixture of methane and ethane with a total volume of 25.0 cm^3 required 65.0 cm^3 of oxygen for complete combustion.

All gas volumes were measured at the same temperature and pressure.
Determine the percentage, by volume, of methane in the mixture.

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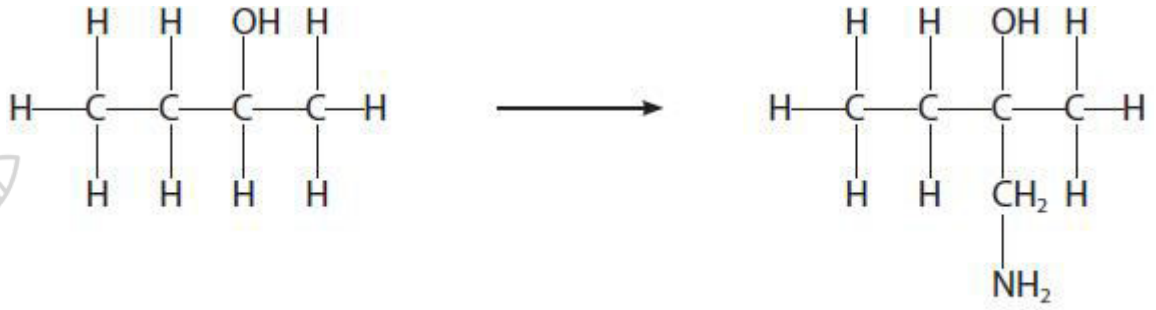
(4)

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(b) Devise a reaction scheme to convert butan-2-ol into 1-amino-2-methylbutan-2-ol in **three** steps.

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1-amino-2-methylbutan-2-ol

Include the reagents, conditions and equations for each of the steps.

For any oxidation or reduction use [O] and [H] respectively.

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(6)

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(Total for question = 10 marks)

Q7.

What is an organic product of the reaction between ethylamine and chloroethane?

☒ **A** NH_2CH_2COCl

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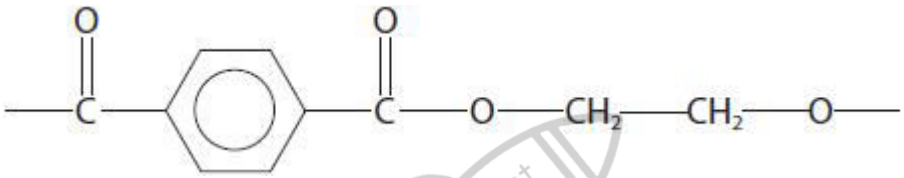
☒ **B** $C_2H_5NH_4^+Cl^-$
☒ **C** $C_2H_5NHCOC_2H_5$
☒ **D** $(C_2H_5)_2NH_2^+Cl^-$

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which pair of monomers can form this polymer?



	Monomer 1	Monomer 2
☒ **A**	HOC_6H_4OH	$HOOCCH_2CH_2COOH$
☒ **B**	$HOCH_2C_6H_4CH_2OH$	$HOOCCH_2CH_2COOH$
☒ **C**	$HOCH_2C_6H_4COOH$	$HOCH_2CH_2COOH$
☒ **D**	$HOOCC_6H_4COOH$	$HOCH_2CH_2OH$

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

Q9.

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Iron Chemistry

Iron is a typical transition metal. Due to the similar energies of the 3d and 4s electrons, iron forms compounds in a number of oxidation states. Iron(II) and iron(III) are the most common oxidation states, and iron(III) is the most stable.

Iron ions form many complexes, including that in haemoglobin which is responsible for oxygen transport in the blood of most vertebrates. The haemoglobin-iron complex with oxygen is responsible for the red colour of blood.

Iron(III) ions may be detected in solution by the addition of thioglycolic acid ($HSCH_2COOH$). All the water ligands of the iron(III) ion are replaced giving a complex with an intense red colour which can be detected in very low concentrations.

The complexes of iron(II) and iron(III) usually have a coordination number of six and are octahedral but the chloro complexes have a coordination number of four and are tetrahedral.

Iron and its compounds can act as catalysts. The element catalyses the Haber process, acting as a typical heterogeneous catalyst. However, the compounds and complexes of iron are usually homogeneous catalysts.

(a) Explain, in terms of electronic structure, why iron(III) compounds are more stable than iron(II) compounds.

(2)

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(b) The third ionisation energy of iron is 2958 kJ mol^{-1}.

(i) Write the equation for the third ionisation energy of iron.
Include state symbols.

(1)

(ii) Explain how **stable** iron(III) ions can be formed from iron(II) ions in aqueous solution.
Refer to the relevant energy changes of these ions only.

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(3)

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(c) Invertebrates use a copper complex, haemocyanin, to transport oxygen.
Blue oxyhaemocyanin gives invertebrate blood its characteristic colour.

Explain why oxyhaemocyanin and oxyhaemoglobin have different colours.

(3)

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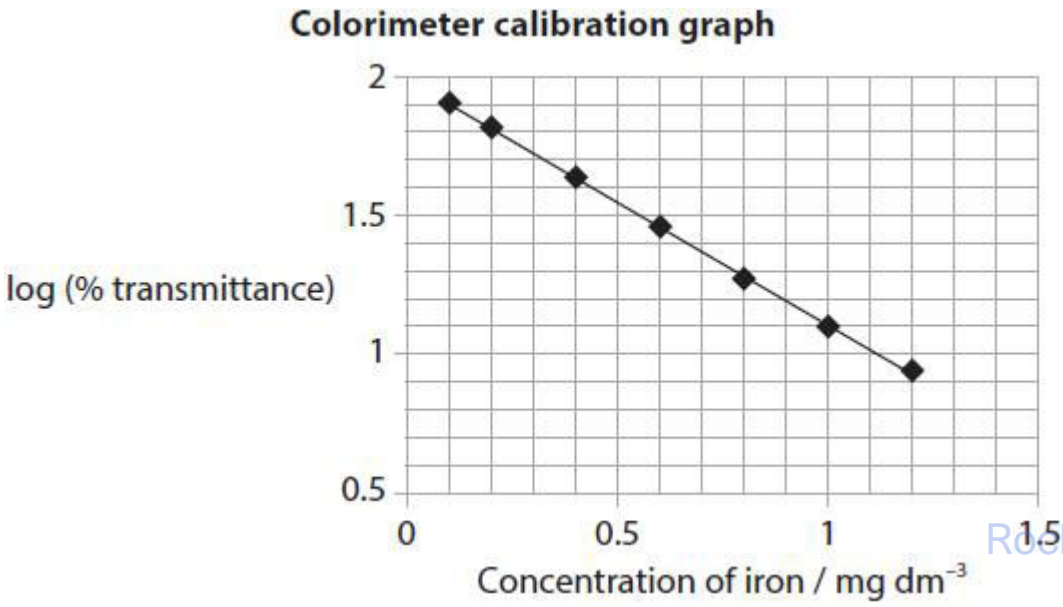
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(d) The presence of iron in sodium carbonate can affect its properties; the higher the quality of the sodium carbonate, the
lower the proportion of iron.

The proportion of iron in a laboratory grade anhydrous sodium carbonate was listed as less than 20 ppm by mass.
In an experiment to check this specification, 20 g of the sodium carbonate was dissolved in sulfuric acid, and thioglycolic acid added in excess to form the iron(III) thioglycolic acid complex, $Fe(HSCH_2COOH)_3^{3+}$ The solution was made up to 500 cm^3 in a volumetric flask and thoroughly mixed.

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The transmittance of the resulting solution was determined using a colorimeter and found to be 39.8 %.

(i) Using the calibration graph, determine whether or not the iron concentration in this sample of sodium carbonate

meets the stated specification.

(4)

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(ii) Suggest what type of ligand thioglycolic acid is in the iron(III) thioglycolic acid complex. Justify your answer.

(2)

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(e) Iodide ions are oxidised to iodine by peroxodisulfate ions.

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$$2I^-(aq) + S_2O_8^{2-}(aq) \rightarrow I_2(aq) + 2SO_4^{2-}(aq)$$

Iron(II) ions act as a homogeneous catalyst for this reaction.

(i) State why the catalyst is described as 'homogeneous'.

(1)

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(ii) Write two equations to show how iron(II) ions catalyse this oxidation.

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State symbols are not required.

(2)

(iii) Suggest how iron(II) ions lower the activation energy of this reaction.

(1)

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(f) Give a possible reason why the chloro complexes of iron ions have a coordination number of four rather than six.

(1)

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(Total for question = 20 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Aqueous sodium hydroxide was added to aqueous iron(II) sulfate and the mixture allowed to stand.

What would be observed?

	Observations	
	Immediately after adding sodium hydroxide	After standing
☐ **A**	brown precipitate	no change
☐ **B**	green precipitate	no change
☐ **C**	brown precipitate	precipitate turns green
☐ **D**	green precipitate	precipitate turns brown

(Total for question = 1 mark)

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Q11.

A yellow crystalline solid **E** dissolved in distilled water to give a yellow solution. Addition of dilute sulfuric acid to this solution produced an orange solution **F**. Warming **F** with ethanol resulted in a green solution **G**, and the formation of ethanal.

A standard cell was set up using solutions of **F** and **G** for the right-hand electrode and ethanol and ethanal for the left-hand electrode. $E^{\ominus}_{cell}$ was found to be +1.94 V.

$$CH_3CHO(aq) + 2H^+(aq) + 2e^- \rightleftharpoons C_2H_5OH(aq) \quad E^{\ominus} = -0.61\ V$$

(a) Deduce the formulae of the ions responsible for the colours of **F** and **G**, using the standard electrode potential and $E^{\ominus}$ given, and the values in the Data Booklet.

(2)

.....

.....

(b) Write the overall equation for the reaction in the cell.
State symbols are not required.

(2)

(c) Write the ionic equation for the reaction of the aqueous solution of **E** with dilute sulfuric acid. State symbols are not required.

(1)

(Total for question = 5 marks)

Q12.

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This question is about titanium(III) chloride, $TiCl_3$.

(a) Titanium(III) chloride is used as a catalyst in the production of poly(propene).

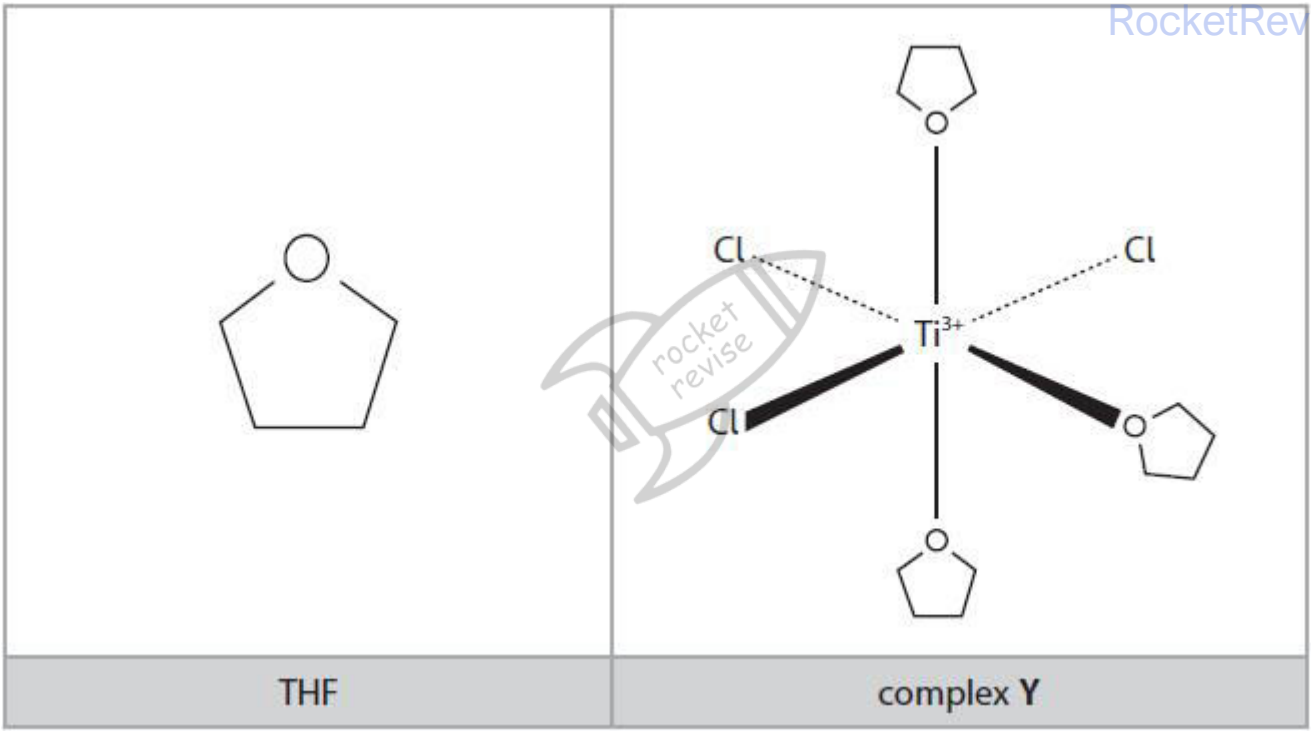
State the property of transition metals, such as titanium, that makes their **compounds** effective catalysts.

(1)

.....

.....

(b) When dissolved in tetrahydrofuran (THF), titanium(III) chloride forms a blue solution containing complex **Y**.



(i) THF acts as a monodentate ligand in complex **Y**.
State the meaning of the terms **monodentate** and **ligand**.

(2)

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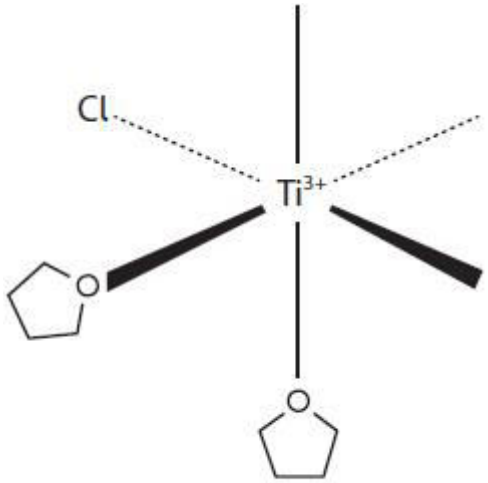
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(ii) Complex **Z** is a stereoisomer of complex **Y**.
Complete the diagram to show the arrangement of the ligands in complex **Z**.

(1)

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complex **Z**

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(c) A student determines the change in oxidation number of **nitrogen** when a solution of magnesium nitrate, $Mg(NO_3)_2$, is titrated with aqueous titanium(III) chloride.

Procedure

Step **1**

Using a volumetric flask, prepare 100.0 cm^3 of an aqueous solution containing 0.750 g of solid $Mg(NO_3)_2 \cdot 6H_2O$.

Step **2**

Pipette 10.0 cm^3 of the solution from Step **1** into a conical flask and add a few drops of alizarin indicator.

Add 2 cm^3 of concentrated hydrochloric acid and heat the mixture.

Step **3**

Fill a burette with 0.0850 mol dm^{-3} aqueous titanium(III) chloride and titrate the contents of the conical flask from Step **2** while continuing to heat the mixture.

During the titration, Ti^{3+} ions are oxidised to TiO^{2+} ions

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$$Ti^{3+} + H_2O \rightarrow TiO^{2+} + 2H^+ + e^-$$

Alizarin indicator is green in the presence of aqueous Ti^{3+} and yellow in the presence of aqueous TiO^{2+}.

The end-point of the titration is reached on the addition of 20.70 cm^3 of aqueous titanium(III) chloride.

(i) State the **colour** change that would be observed at the end-point of the titration.

(1)

From ..... to .....

(ii) Use the results to determine the **final** oxidation state of nitrogen in the titration.

You **must** show your working.

[M_r value: $Mg(NO_3)_2 \cdot 6H_2O$ = 256.3]

(5)

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(iii) Write the overall **ionic** equation for the titration reaction using your answer to (c)(ii) and the relevant half-equations from the list below.

(2)

$TiO^{2+} + 2H^+ + e^-$	$\rightleftharpoons$	$Ti^{3+} + H_2O$	$E^{\ominus} = +0.10\,V$
$NO_3^- + 2H^+ + e^-$	$\rightleftharpoons$	$NO_2 + H_2O$	$E^{\ominus} = +0.80\,V$
$NO_3^- + 3H^+ + 2e^-$	$\rightleftharpoons$	$HNO_2 + H_2O$	$E^{\ominus} = +0.94\,V$
$NO_3^- + 4H^+ + 3e^-$	$\rightleftharpoons$	$NO + 2H_2O$	$E^{\ominus} = +0.96\,V$

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(iv) Use the electrode potential data provided to calculate $E^{\ominus}_{cell}$ for the overall titration reaction.

(1)

(v) Suggest why the contents of the conical flask are heated.

(1)

.....

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.....

.....

*(vi) The student's teacher said,

"*As $TiCl_3$ is blue and TiO^{2+} ions are colourless in aqueous solution, the titration can be carried out* ***without*** *an alizarin indicator*."

Assess the teacher's statement.

In your answer you should

- identify, by name or formula, a coloured complex ion expected to be present when $TiCl_3$ dissolves in water
- explain how the colour of this complex ion arises
- suggest why the titration may be more accurate **with** an alizarin indicator.

(6)

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(Total for question = 20 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Copper(II) ions form a complex with 1,2-diaminoethane (symbol 'en') with the formula $Cu(en)_3^{2+}$.

What type of ligand is 1,2-diaminoethane, and what is the coordination number of copper(II) in the complex?

	Type of ligand	Coordination number
☐ A	bidentate	3
☐ B	bidentate	6
☐ C	tridentate	3
☐ D	tridentate	6

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(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When aqueous ammonia is added to an aqueous solution of zinc sulfate, a white precipitate forms which dissolves in excess ammonia to give a colourless solution.

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What types of reaction are occurring?

	Type of reaction	
	Formation of white precipitate	Formation of colourless solution
☐ A	deprotonation	ligand exchange
☐ B	deprotonation	deprotonation
☐ C	ligand exchange	deprotonation
☐ D	ligand exchange	ligand exchange

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(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these has the greatest number of unpaired electrons in each of its atoms?

☐ **A** chromium

☐ **B** iron

☐ **C** manganese

☐ **D** vanadium

(Total for question = 1 mark)

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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The reaction of ethanedioate ions, $C_2O_4^{2-}$, with manganate(VII) ions, MnO_4^- in acidic solution involves autocatalysis.

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

The catalyst in this reaction is

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☒ **A** MnO_4^-

☒ **B** H^+

☒ **C** Mn^{2+}

☒ **D** CO_2

(Total for question = 1 mark)

Q17.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following statements **best** explains carbon monoxide poisoning?

☒ **A** carbon monoxide binds irreversibly to haemoglobin

☒ **B** carbon monoxide forms stronger dative covalent bonds with haemoglobin than oxygen does

☒ **C** the formation of carboxyhaemoglobin leads to a large increase in the entropy of the system

☒ **D** carbon monoxide has a triple bond whereas oxygen has a double bond

(Total for question = 1 mark)

Q18.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Iodide ions are oxidised by peroxodisulfate ions in aqueous solution.

$$2I^- + S_2O_8^{2-} \rightarrow I_2 + 2SO_4^{2-}$$

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This reaction is catalysed by adding Fe^{2+} ions to the solution.

This catalysis is effective because

☒ **A** Fe^{2+} reacts with iodide ions and with peroxodisulfate ions

☒ **B** Fe^{2+} has many electrons in its outermost subshells

☒ **C** Fe^{2+} has many active sites on which the reaction can occur

☒ **D** Fe^{2+} is readily oxidised to Fe^{3+} which is then reduced to Fe^{2+}

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(Total for question = 1 mark)

Q19.

* Compare and contrast the mechanism of the action of platinum as a catalyst in the removal of pollutants from car engine exhaust fumes with that of vanadium(V) oxide as a catalyst in the Contact Process for the manufacture of sulfuric acid.

General definitions of catalysts are **not** required.

(6)

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(Total for question = 6 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Iodide ions, I^-, are oxidised by peroxodisulfate(VI) ions, $S_2O_8^{2-}$.

$$2I^-(aq) + S_2O_8^{2-}(aq) \rightarrow I_2(aq) + 2SO_4^{2-}(aq)$$

Which of the following statements is **true** of this reaction?

- ☐ **A** both $Fe^{2+}(aq)$ and $Fe^{3+}(aq)$ catalyse the reaction
- ☐ **B** $Fe^{2+}(aq)$ catalyses the reaction but $Fe^{3+}(aq)$ does not
- ☐ **C** $Fe^{3+}(aq)$ catalyses the reaction but $Fe^{2+}(aq)$ does not
- ☐ **D** neither $Fe^{2+}(aq)$ nor $Fe^{3+}(aq)$ catalyses the reaction

(Total for question = 1 mark)

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Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following is **not** true of the reactions occurring in the catalytic converter fitted to a car exhaust?

- ☐ **A** they involve heterogeneous catalysis
- ☐ **B** carbon monoxide is adsorbed onto the surface of the catalyst
- ☐ **C** nitrogen is desorbed from the surface of the catalyst
- ☐ **D** the products cause no harm to the environment

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(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Cobalt chloride is used as a test for the presence of water.

This test depends on the fact that

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☐ **A** anhydrous cobalt(II) chloride is blue and hydrated cobalt(II) chloride is pink

☐ **B** anhydrous cobalt(II) chloride is pink and hydrated cobalt(II) chloride is blue

☐ **C** cobalt(II) chloride is blue and cobalt(III) chloride is pink

☐ **D** cobalt(II) chloride is pink and cobalt(III) chloride is blue

(Total for question = 1 mark)

Q23.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Platinum forms a complex with the formula $Pt(NH_3)_2Cl_2$ and chromium forms a complex with the formula $CrCl_4^-$.

(a) What are the shapes of these complexes?

(1)

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☐ **A** both complexes are square planar

☐ **B** both complexes are tetrahedral

☐ **C** $Pt(NH_3)_2Cl_2$ is tetrahedral and $CrCl_4^-$ is square planar

☐ **D** $Pt(NH_3)_2Cl_2$ is square planar and $CrCl_4^-$ is tetrahedral

(b) What is the bonding between the ligands and the central atom in these complexes?

(1)

☐ **A** the bonding in both complexes is ionic

☐ **B** the bonding in both complexes is dative covalent

☐ **C** the bonding in $Pt(NH_3)_2Cl_2$ is dative covalent and in $CrCl_4^-$ is ionic

☐ **D** the bonding in $Pt(NH_3)_2Cl_2$ is ionic and in $CrCl_4^-$ is dative covalent

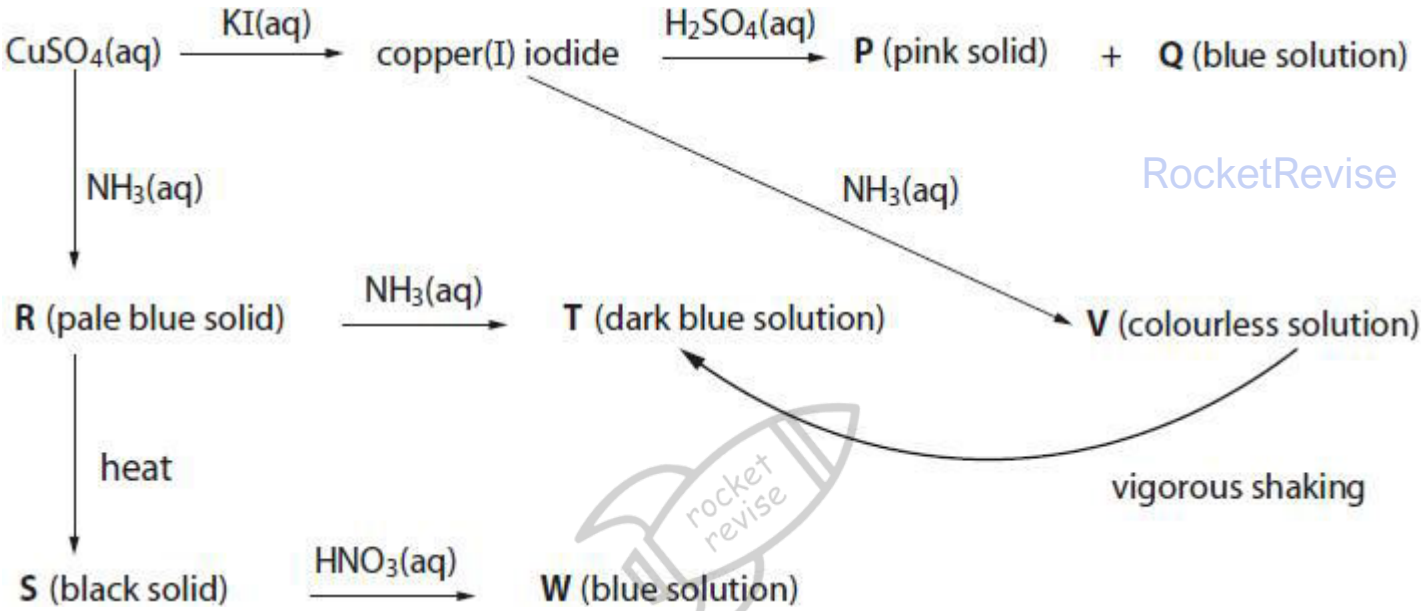
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(Total for question = 2 marks)

Q24.

The diagram summarises some reactions of copper compounds.



(a) Identify, by name (including the oxidation state) or formula, the species in the sequence that contain copper.

(7)

P .....

Q .....

R .....

S .....

T .....

V .....

W .....

(b) **T** and **V** are the same type of chemical species.

(i) Name this type of chemical species.

(1)

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(ii) Explain why **T** is coloured while **V** is colourless.
A detailed explanation of the fact that **T** is coloured is **not** required.

(3)

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(iii) Suggest an explanation for the change of **V** into **T** on shaking.

(2)

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(c) The reaction between copper(I) iodide and sulfuric acid is a disproportionation.

(i) Write the **ionic** equation for this disproportionation reaction.
State symbols are not required.

(1)

(ii) Show that the reaction in (c)(i) is thermodynamically feasible.
Use the standard electrode potentials of the relevant half-cells from the Data Booklet.

(2)

(d) The rare mineral mitscherlichite has the chemical formula $K_2CuCl_4 \cdot nH_2O$.

4.26 g of mitscherlichite was dissolved in distilled water and the solution made up to 250.0 cm^3.
Excess potassium iodide solution was added to a 25.0 cm^3 portion of this solution and the iodine formed was titrated against a solution of sodium thiosulfate with a concentration of 0.0500 mol dm^{-3}.
This procedure was repeated until concordant results were obtained.
The mean accurate titre was 26.65 cm^3.
The equations for the reactions are

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$$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$$

$$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$$

Calculate the value of n, the number of moles of water of crystallisation per mole of mitscherlichite.

(6)

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(Total for question = 22 marks)

Q25.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Diamminecopper(I) ions are **not** coloured because

☐ **A** the d orbitals in copper(I) cannot be split

☐ **B** the energy difference between the split d orbitals is outside the visible region of the spectrum

☐ **C** d—d transitions are not possible because the d orbitals are fully occupied

☐ **D** copper(I) complexes are readily oxidised

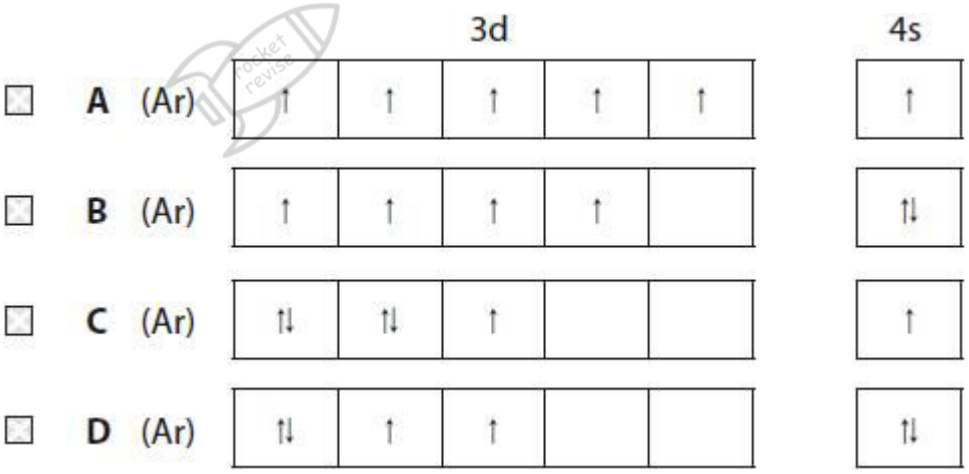
(Total for question = 1 mark)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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What is the electronic configuration of a chromium atom?



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(Total for question = 1 mark)

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some nickel(II) complex ions are formed by the addition of complexing agents to nickel(II) ions, $[Ni(H_2O)_6]^{2+}$, in aqueous solution.

On formation, which of these leads to the **most** positive increase in ΔS_{system}?

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- ☐ **A** $[NiCl_4]^{2-}$
- ☐ **B** $[Ni(EDTA)]^{2-}$
- ☐ **C** $[Ni(C_2O_4)_2]^{2-}$
- ☐ **D** $[Ni(H_2NCH_2CH_2NH_2)_3]^{2+}$

(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A ligand must be an

- ☐ **A** electron-pair donor
- ☐ **B** electron-pair donor and negatively charged
- ☐ **C** electron-pair acceptor
- ☐ **D** electron-pair acceptor and negatively charged

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(Total for question = 1 mark)

Q29.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Nickel is classified as a transition metal. This is because nickel

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☒ **A** is a d block element

☒ **B** has partially filled d orbitals

☒ **C** forms stable ions with partially filled d orbitals

☒ **D** forms stable compounds in which it has different oxidation states

(Total for question = 1 mark)

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Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Aqueous ammonia is added drop by drop to a solution of cobalt(II) chloride, $CoCl_2(aq)$, until in excess.

What would be the sequence of observations?

☒ **A** blue solution → pink precipitate → dark blue solution

☒ **B** pink solution → blue precipitate

☒ **C** blue solution → pink precipitate

☒ **D** pink solution → blue precipitate → yellow-brown solution

(Total for question = 1 mark)

Q31.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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This question is about catalysts.

(a) Some standard electrode potentials are shown.

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Right-hand electrode system	$E^{\ominus}$/V
$Cu^{2+} + e^- \rightleftharpoons Cu^+$	+0.15
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.54
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36
$S_2O_8^{2-} + 2e^- \rightleftharpoons 2SO_4^{2-}$	+2.01

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Which of these ions is most likely to catalyse the reaction between $S_2O_8^{2-}$ and I^-?

$$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$$

(1)

- ☒ **A** Cl^-
- ☒ **B** Fe^{2+}
- ☒ **C** Cu^{2+}
- ☒ **D** Cu^+

(b) Which term best describes the type of catalyst for the reaction in (a)?

(1)

- ☒ **A** autocatalyst
- ☒ **B** biocatalyst
- ☒ **C** heterogeneous
- ☒ **D** homogeneous

(c) Which substance is manufactured in a process involving a reaction catalysed by vanadium(V) oxide?

(1)

- ☒ **A** ammonia
- ☒ **B** nitric acid
- ☒ **C** sodium hydroxide
- ☒ **D** sulfuric acid

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(Total for question = 3 marks)

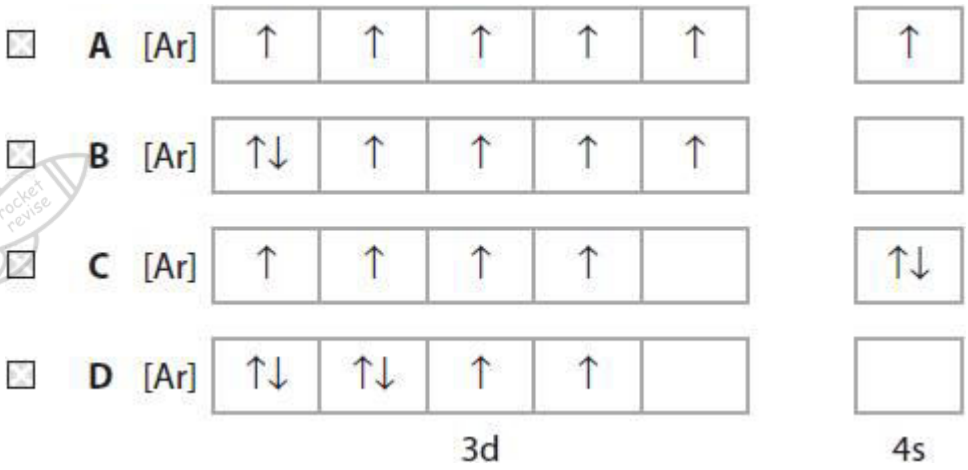
Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the electronic configuration of the Fe^{2+} ion?

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(Total for question = 1 mark)

Q33.

Gilding Metal

Gilding metal is a type of brass alloy that consists of copper and a small amount of zinc, ranging from 5% to 11% by mass. Copper is very malleable and is hardened by the addition of zinc.

Gilding metal is much less susceptible to cracking due to corrosion than brasses with a higher percentage of zinc.

It has a warm, golden colour and can be used to coat materials using electrolysis. It is also used to make test pieces in jewellery manufacture because it has similar properties to silver but is less expensive.

The proportions of copper and zinc determine the exact properties of the gilding metal and can be determined by chemical analysis.

(a) 2.72 g of a type of brass is dissolved in excess concentrated nitric acid, forming a solution containing both Cu^{2+} and Zn^{2+} ions.

A solution containing hydrogensulfate(IV) ions, HSO_3^-, is then added.

$$2Cu^{2+}(aq) + HSO_3^-(aq) + H_2O(l) \rightarrow 2Cu^+(aq) + HSO_4^-(aq) + 2H^+(aq)$$

The addition of ammonium thiocyanate, NH4SCN, gives a precipitate of copper(I) thiocyanate, CuSCN.

$$Cu^+(aq) + SCN^-(aq) \rightarrow CuSCN(s)$$

The precipitate of copper(I) thiocyanate is collected, dried and found to have a mass of 4.69 g.

(i) Determine whether or not this type of brass is a gilding metal, by calculating its percentage by mass of copper.

(4)

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(ii) Explain, by considering both thermodynamic and kinetic factors, why HSO_3^- reduces Cu^{2+} to Cu^+ but does not then reduce Cu^+ to Cu.

Use the data in the table.

(3)

Right-hand electrode system	$E^{\ominus}$/V
$Cu^{2+} + e^- \rightleftharpoons Cu^+$	+0.15
$HSO_4^- + 2H^+ + 2e^- \rightleftharpoons HSO_3^- + H_2O$	+0.17
$Cu^+ + e^- \rightleftharpoons Cu$	+0.52

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(b) After the copper(I) thiocyanate is precipitated, Zn^{2+} ions remain in solution.

A student suggested that these Zn^{2+} ions can be precipitated by adding a large **excess** of aqueous sodium hydroxide.

Comment on this suggestion by describing the reactions that take place as a large **excess** of aqueous sodium hydroxide is **gradually** added.

(4)

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(c) Suggest why gilding metals are less malleable than pure copper, by considering their structure.

(2)

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(d) Zinc and copper are also used in electrochemical cells.

(i) Draw a labelled diagram of the apparatus used to measure the emf of a cell with copper and zinc electrodes under standard conditions.

(3)

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(ii) The Nernst equation describes the relationship between the concentration of metal ions in a half-cell and its electrode potential.

$$E = E^{\ominus} + \frac{0.0260}{z} \times \ln[\text{ion}]$$

E = electrode potential under non-standard concentrations
z = the number of positive charges on the metal ion
A cell is set up with Cu^{2+} ions of concentration 1.00 mol dm^{-3} and Zn^{2+} ions of unknown concentration. The emf of the cell is +1.09 V.
Calculate the concentration of the zinc ions.
Use the data on page 10 of the Data Booklet.

(3)

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(Total for question = 19 marks)

Q34.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The formulae of four ions are shown.

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Formula of ion
CrO_4^{2-}
AlO_2^-
$[Fe(CN)_6]^{4-}$
$[CrCl_2(H_2O)_4]^+$

How many of these ions contain a metal with an oxidation number of +3?

- ☒ **A** one
- ☒ **B** two
- ☒ **C** three
- ☒ **D** four

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(Total for question = 1 mark)

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When oxygen binds to the haem group in haemoglobin, each oxygen molecule

- ☒ **A** bonds reversibly to an iron(II) ion
- ☒ **B** bonds irreversibly to an iron(II) ion
- ☒ **C** replaces an iron(II) ion in a reversible reaction
- ☒ **D** replaces an iron(II) ion in an irreversible reaction

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(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q36.

Many oxides of transition metals are used as coloured pigments.

(a) Viridian is a blue-green pigment with the formula $M_2O_3 \cdot 2H_2O$; M is not the symbol of the element.

When a sample of viridian is heated until all the water of crystallisation is removed, the mass is reduced by 19.15%.
Identify element M.

(4)

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(b) Cobalt(II) oxide is used in the ceramics industry as an additive to produce blue glazes and enamels. Cobalt(II) oxide dissolves in sulfuric acid to give a pink aqueous solution of cobalt(II) sulfate. When concentrated hydrochloric acid is added to aqueous cobalt(II) sulfate, a dark blue solution forms.

(i) Name the type of reaction that occurs when concentrated hydrochloric acid is added to aqueous cobalt(II) sulfate.

(1)

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(ii) Write an **ionic** equation for the reaction that occurs when concentrated hydrochloric acid is added to aqueous cobalt(II) sulfate.
State symbols are not required.

(2)

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(iii) Explain why the shape of the complex ion changes when concentrated hydrochloric acid is added.

(2)

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(Total for question = 9 marks)

Q37.

Describe the reactions of separate samples of aqueous cobalt(II) sulfate with aqueous sodium hydroxide, excess aqueous ammonia and concentrated hydrochloric acid.

For each reaction, link your description to an appropriate equation.
State symbols are not required.

(Total for question = 6 marks)

Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Excess aqueous sodium thiosulfate is added to an aqueous solution of ammonium vanadate(V).

What colour is the mixture when the reaction is complete?

Refer to page 10 of the Data Booklet.

(1)

- ☐ **A** yellow
- ☐ **B** blue
- ☐ **C** green
- ☐ **D** violet

Use this space for rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

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Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reagent, when added to aqueous sodium dichromate(VI), $Na_2Cr_2O_7(aq)$, causes a shift in equilibrium resulting in the formation of a yellow solution?

- ☐ **A** NaOH(aq)
- ☐ **B** HCl(aq)
- ☐ **C** Zn(s)
- ☐ **D** $H_2O_2(aq)$

(Total for question = 1 mark)

Q40.

This question is about the chemistry of vanadium.

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The standard electrode potentials of some vanadium species are shown.

Electrode system	$E^{\ominus}$ / V
$V^{2+}(aq) + 2e^- \rightleftharpoons V(s)$	−1.18
$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	−0.26
$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+0.34
$VO_3^-(aq) + 4H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + 2H_2O(l)$	+1.00

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(a) Explain the highest stable oxidation state formed by vanadium, by referring to its electronic configuration.

(3)

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(b) A student suggests that the ion VO^{2+} may be converted into V^{3+} using sodium thiosulfate, $Na_2S_2O_3$, with no other vanadium species being formed by reduction.

(i) Justify the use of sodium thiosulfate for this reaction by writing the relevant equations and calculating their $E^{\ominus}_{cell}$ values.

Use the standard electrode potentials given in the table and values from your Data Booklet.

State symbols are not required in the equations.

(4)

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(ii) Explain why nickel, Ni, is **not** a suitable reagent to convert VO^{2+} into V^{3+}, with no other vanadium species being formed.

(2)

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(c) Most vanadium produced is used to make a steel alloy called ferrovanadium.

The vanadium content of ferrovanadium may be determined by a titration method.

Procedure

- The sample of ferrovanadium is dissolved in chloric(V) acid. The vanadium species formed is VO_3^- .
- The resulting solution is transferred to a 250.0 cm^3 volumetric flask, washings added and the solution made up to the mark with distilled water and mixed.
- Using a pipette, 25.0 cm^3 of the solution is transferred to a conical flask and 25.0 cm^3 of a 0.250 mol dm^{-3} solution of iron(II) sulfate, $FeSO_4$, is added. The iron(II) ions react with the VO_3^- ions:

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$$VO_3^- + 4H^+ + Fe^{2+} \leftrightarrow VO^{2+} + 2H_2O + Fe^{3+}$$

- The resulting solution is titrated against potassium manganate(VII) to determine the amount of iron(II) ions remaining.

$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

In an experiment, the mass of ferrovanadium used was 4.87 g, the concentration of potassium manganate(VII) was 0.0195 mol dm^{-3} and a mean titre of 22.50 cm^3 was obtained.

(i) Give the colour of the solution at the end-point of the titration.

(1)

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(ii) Suggest why the VO^{2+} ions formed do **not** affect the titration.

(2)

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(iii) Calculate the percentage by mass of vanadium in the ferrovanadium.

(7)

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(d) In the manufacture of sulfuric acid, vanadium(V) oxide, V_2O_5, is the catalyst used in the conversion of sulfur dioxide to sulfur trioxide:

$$2SO_2 + O_2 \leftrightarrow 2SO_3$$

Write **two** equations to show a possible mechanism for this reaction.

State symbols are not required.

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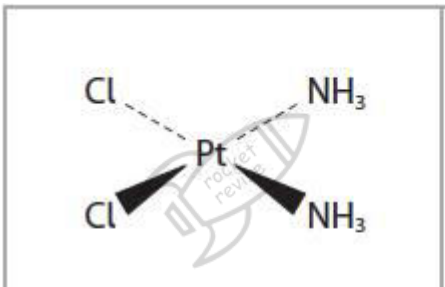
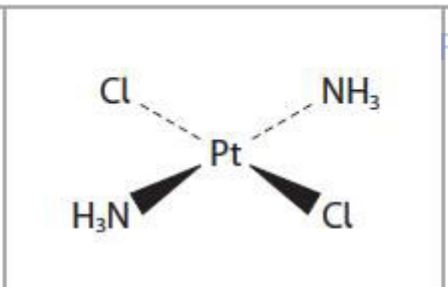
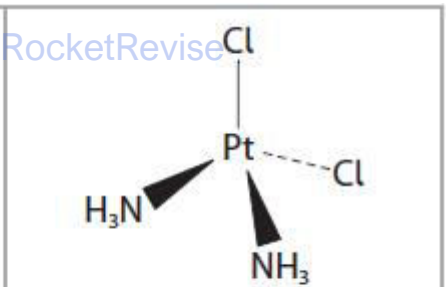
(2)

(Total for question = 21 marks)

Q41.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Platinum forms a complex with the formula $Pt(NH_3)_2Cl_2$ which is used in cancer treatment. Three possible structures of this complex are shown.

		
X	Y	Z

The complex used in cancer treatment contains

- ☒ **A** structure X only
- ☒ **B** structure Y only
- ☒ **C** structure Z only

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☒ **D** an equimolar mixture of structures X and Y only

(Total for question = 1 mark)

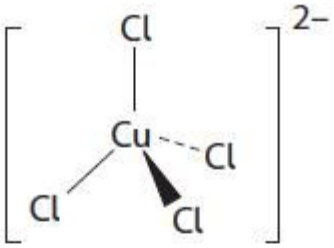
Q42.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

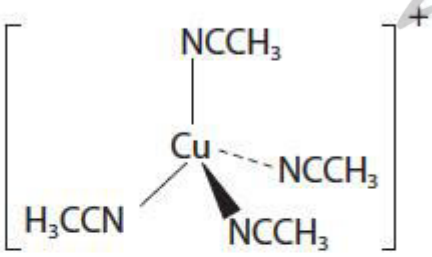
Which transition metal complex is commonly used as a treatment for cancer?

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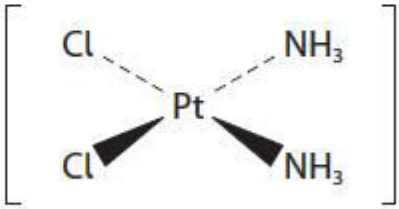
☒ **A**



☒ **B**

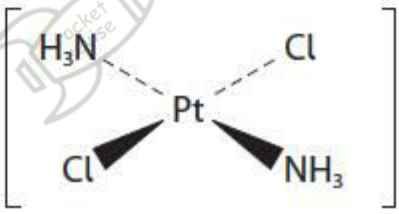


☒ **C**



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☒ **D**



(Total for question = 1 mark)

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Q43.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The sequence shown is the mechanism for a reaction in aqueous solution.

Step 1 $Ag^{+} + Ce^{4+} \rightarrow Ag^{2+} + Ce^{3+}$
Step 2 $Ti^{+} + Ag^{2+} \rightarrow Ti^{2+} + Ag^{+}$
Step 3 $Ti^{2+} + Ce^{4+} \rightarrow Ti^{3+} + Ce^{3+}$

In the **overall** reaction

- ☐ **A** the oxidation of Ag+ is catalysed by Ce^{4+} ions
- ☐ **B** the oxidation of Ag+ is catalysed by Ti^{2+} ions
- ☐ **C** the oxidation of Ti+ is catalysed by Ag^{+} ions
- ☐ **D** the oxidation of Ti+ is catalysed by Ag^{2+} ions

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(Total for question = 1 mark)

Q44.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The element zinc is **not** classified as a transition metal. This is because

- ☐ **A** the 3d subshell of a zinc atom is full
- ☐ **B** zinc only forms one stable ion
- ☐ **C** the only stable zinc ion has the electronic configuration [Ar] $3d^{10}$
- ☐ **D** neither zinc nor zinc ions show catalytic properties

(Total for question = 1 mark)

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Q45.

This question is about transition metal compounds and their quantitative analysis.

(a) Potassium dichromate(VI), K_2Cr_2O7, is present in very small amounts in cement, to help increase the time for the cement to set.

A 50.0 g sample of cement was washed with portions of deionised water to dissolve the potassium dichromate(VI). Any insoluble residues were removed by filtration and the filtrate was transferred to a volumetric flask. The volume was made up to 100.0 cm^3 , using 2 mol dm^{-3} sulfuric acid.
50.0 cm^3 of this solution was transferred to a conical flask and titrated with a solution of ammonium iron(II) sulfate, $(NH_4)2Fe(SO_4)_2(aq)$, of concentration 3.24×10^{-4} mol dm^{-3} .
The indicator *N*-phenylanthranilic acid was used, which gave an intense red-violet colour at the end-point.
The mean titre of ammonium iron(II) sulfate was 10.90 cm^3 .

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The ionic equation for the redox reaction in the titration is shown.

$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(l) + 6Fe^{3+}(aq)$

(i) State the colour of each chromium species in the reaction.

(1)

$Cr_2O_7^{2-}$ .....

Cr^{3+} .....

(ii) Suggest a reason why an indicator is needed in this titration.

(1)

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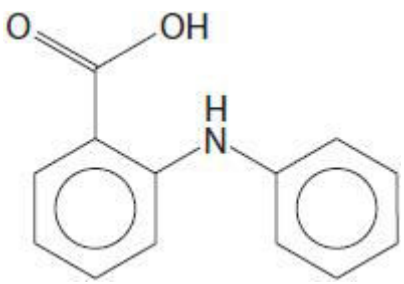
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(iii) Calculate the percentage by mass of potassium dichromate(VI) in the cement sample.

(5)

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(b) N-phenylanthranilic acid has the structure shown.



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The solution used as an indicator was prepared by mixing 100 mg of this acid in 5 cm^3 of sodium hydroxide solution, NaOH(aq).
The mixture was then diluted to 100 cm^3 with deionised water.
Explain why the *N*-phenylanthranilic acid is added to the sodium hydroxide solution before it is mixed with water in the preparation of this solution.

(2)

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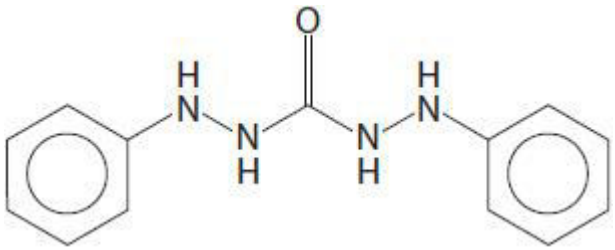
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(c) The concentration of chromium(VI) in aqueous solution may also be determined using a colorimeter.

On adding 1,5-diphenylcarbazide, DPC, to a solution of chromium(VI) ions, an intensely coloured octahedral complex forms.

The formula of the complex is $Cr(DPC)_3^{6+}$.

(i) The structure of DPC is shown.



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Describe how DPC is able to act as a bidentate ligand, using your diagram to show the atoms involved.

(3)

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(ii) The intense colour of this complex is due to the transfer of electrons from the ligand to the chromium(VI) ion.

Suggest a possible reason why the colour is **not** due to the transfer of electrons between split d-orbitals in the ion.

Refer to the electronic configuration of the chromium(VI) ion.

(1)

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(d) The concentration of nickel(II) ions, $Ni^{2+}(aq)$, can be determined by forming a complex with the ligand dimethylglyoxime, $C_4H_8N_2O_2$.

$$[Ni(H_2O)_6]^{2+} + 2C_4H_8N_2O_2 + 2OH^- \rightarrow [Ni(C_4H_7N_2O_2)_2(H_2O)_2] + 6H_2O$$

Explain why the formation of the dimethylglyoxime complex is favoured, in terms of entropy.

(2)

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(Total for question = 15 marks)

Q46.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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This question is about transition metal complexes.

(a) The bonding within the complex $[Cu(NH_3)_4(H_2O)_2]^{2+}$ is

(1)

- ☐ **A** covalent, dative covalent and ionic
- ☐ **B** covalent and dative covalent only
- ☐ **C** covalent only
- ☐ **D** dative covalent only

(b) Which complex is tetrahedral?

(1)

- ☐ **A** $[Pt(NH_3)_2Cl_2]$
- ☐ **B** $[Cu(H_2O)_4(OH)_2]$
- ☐ **C** $[Cu(NH_3)_4(H_2O)_2]^{2+}$
- ☐ **D** $[CoCl_4]^{2-}$

(c) Which complex contains a bidentate ligand?

(1)

- ☐ **A** $[Co(NH_2CH_2CH_2NHCH_2CH_2NH_2)_2]^{3+}$
- ☐ **B** $[Cu(NH_3)_4(H_2O)_2]^{2+}$
- ☐ **C** $[Ni(NH_2CH_2CH_2NH_2)_3]^{2+}$
- ☐ **D** $[Mn(EDTA)]^{2-}$

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(Total for question = 3 marks)

Q47.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which ion is **not** blue?

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- ☐ **A** VO^{2+}
- ☐ **B** $[Co(H_2O)_6]^{2+}$
- ☐ **C** $[Cr(H_2O)_6]^{2+}$
- ☐ **D** $[Cu(H_2O)_6]^{2+}$

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q48.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which species does the transition metal have the **lowest** oxidation number?

- ☐ **A** $[Cu(NH_3)_2]^+$
- ☐ **B** $[Fe(CN)_6]^{3-}$
- ☐ **C** $MnSO_4$
- ☐ **D** $Ni(CO)_4$

(Total for question = 1 mark)

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Q49.

Vanadium forms compounds with oxidation states +2, +3, +4 and +5.

(a) State the oxidation number of vanadium in Na_3VO_4.

(1)

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(b) In aqueous solution, vanadium(II) exists as $[V(H_2O)_6]^{2+}$ but vanadium(V) exists as $[VO_2(H_2O)_4]^+$.

Explain why vanadium(V) does **not** exist as $[V(H_2O)_6]^{5+}$.

(3)

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(c) Hydrated potassium vanadium(III) sulfate contains 7.9 % potassium, 10.2 % vanadium, 12.9 % sulfur, 4.8 % hydrogen and 64.2 % oxygen.

Calculate the empirical formula of this compound and hence write the overall formula showing the potassium (K^+), vanadium (V^{3+}) and sulfate (SO_4^{2-}) ions and the water of crystallisation.

(3)

(d) An acidified aqueous solution of potassium manganate(VII) oxidises vanadium(III) ions.

In an experiment, 10.0 cm^3 portions of a solution containing
0.132 mol dm^{-3} vanadium(III) ions are titrated against
0.0200 mol dm^{-3} potassium manganate(VII) in acidic conditions.
The mean titre is 26.40 cm^3 .

(i) Calculate the ratio of the number of moles of manganate(VII) ions, MnO_4^-, to the number of moles of vanadium(III) ions, V^{3+}, in this oxidation.

You **must** show all your working.

(3)

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(ii) Write the equation for the reaction between manganate(VII) ions and vanadium(III) ions in acidic solution.
State symbols are not required.

(2)

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* (e) Vanadium(V) ions, VO_2^+, in acidic solution may be reduced step by step through its oxidation states.

Compare and contrast the use of iron and tin in this reduction process. Include equations and $E^{\ominus}_{cell}$ values for any reactions that occur. State symbols are not required.

(6)

Use only data from these tables.
Vanadium systems

Electrode system	$E^{\ominus}$ / V
$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	−0.26
$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+0.34
$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	+1.00

Reducing agents

Electrode system	$E^{\ominus}$ / V
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.44
$Sn^{2+}(aq) + 2e^- \rightleftharpoons Sn(s)$	−0.14

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(Total for question = 18 marks)

Q50.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which complex contains **only** monodentate ligands?

- ☒ **A** $[Fe(CN)_6]^{4-}$
- ☒ **B** $[Cr(C_2O_4)_3]^{3-}$
- ☒ **C** $[Ni(EDTA)]^{2-}$
- ☒ **D** $[Co(NH_2CH_2CH_2NH_2)2Cl_2]$

(Total for question = 1 mark)

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Q51.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which metal hydroxide precipitate is soluble in **both** excess aqueous ammonia and excess aqueous sodium hydroxide?

- ☒ **A** $Cu(OH)_2$
- ☒ **B** $Fe(OH)_2$
- ☒ **C** $Ni(OH)_2$
- ☒ **D** $Zn(OH)_2$

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(Total for question = 1 mark)

Q52.

This question is about mercury, Hg, and its compounds.

Mercury is a liquid element in the same group of the Periodic Table as zinc.

The electronic configuration of mercury is $[Xe]4f^{14}5d^{10}6s^2$.

(a) Mercury forms compounds in either the +1 or +2 oxidation states.

Explain why mercury is classified as a d-block element but is **not** a transition element.

(3)

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(b) Mercury reacts with nitric acid to form an aqueous solution of $Hg(NO_3)_2$ and nitrogen monoxide gas.

The **unbalanced** equation is shown.

$$Hg(l) + HNO_3(aq) \rightarrow Hg(NO_3)_2(aq) + NO(g) + H_2O(l)$$

(i) Explain, using oxidation numbers, why this is a redox reaction.

(2)

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(ii) Deduce the **ionic** half-equations for this reaction.
State symbols are not required.

(2)

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(iii) Complete the equation for this reaction by adding the stoichiometric coefficients.

(1)

.....$Hg(l)$ + ...$HNO_3(aq)$ → .....$Hg(NO_3)_2(aq)$ + ...$NO(g)$ + ...$H_2O(l)$

(c) Mercury(II) fulminate, $Hg(CNO)_2$, is an explosive.

(i) It is produced in the reaction between $Hg(NO_3)_2$ and ethanol. The other products of the reaction are ethanal and water.
Write the equation for the reaction of one mole of $Hg(NO_3)_2$ with ethanol to form mercury(II) fulminate.
State symbols are not required.

(2)

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(ii) $Hg(CNO)_2$ decomposes as shown.

$$3Hg(CNO)_2(s) \rightarrow Hg(CN)_2(s) + 2Hg(s) + 2CO_2(g) + 2CO(g) + 2N_2(g)$$

Calculate the **total** volume, in cm^3, of gas produced when 1.00 g of $Hg(CNO)_2$ decomposes at room temperature and pressure.

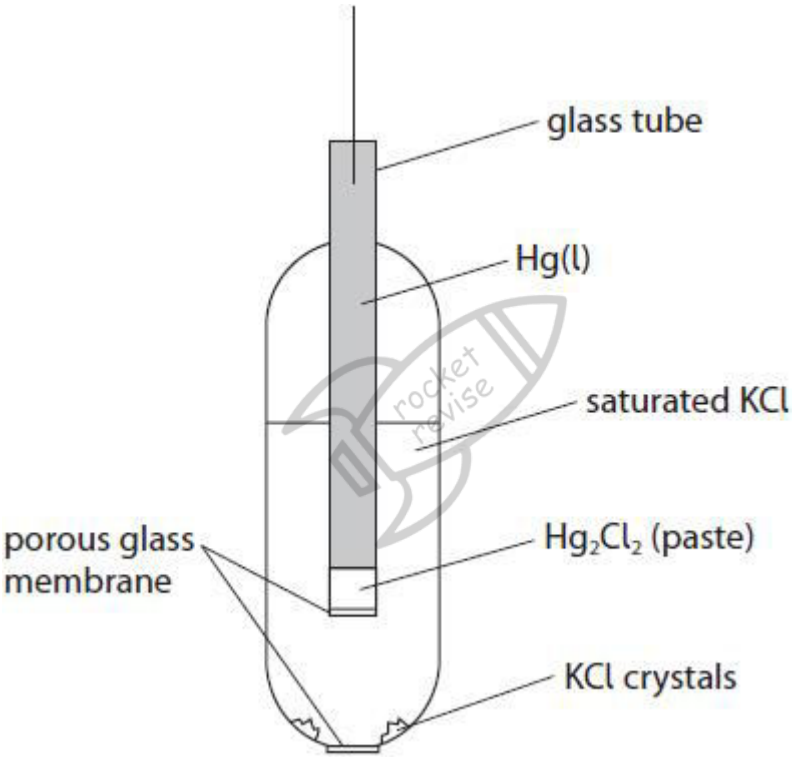
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(d) Mercury(I) chloride, Hg_2Cl_2, is also known as calomel.

A saturated calomel electrode may be used as an alternative to the standard hydrogen electrode.



The half-equation for the calomel electrode is

$$Hg_2Cl_2(s) + 2e^- \rightleftharpoons 2Hg(l) + 2Cl^-(aq)$$

The standard electrode potential of the calomel electrode is $E^{\ominus} = +0.24\,V$.

(i) Suggest why KCl crystals are needed in the outer tube of the electrode.

(1)

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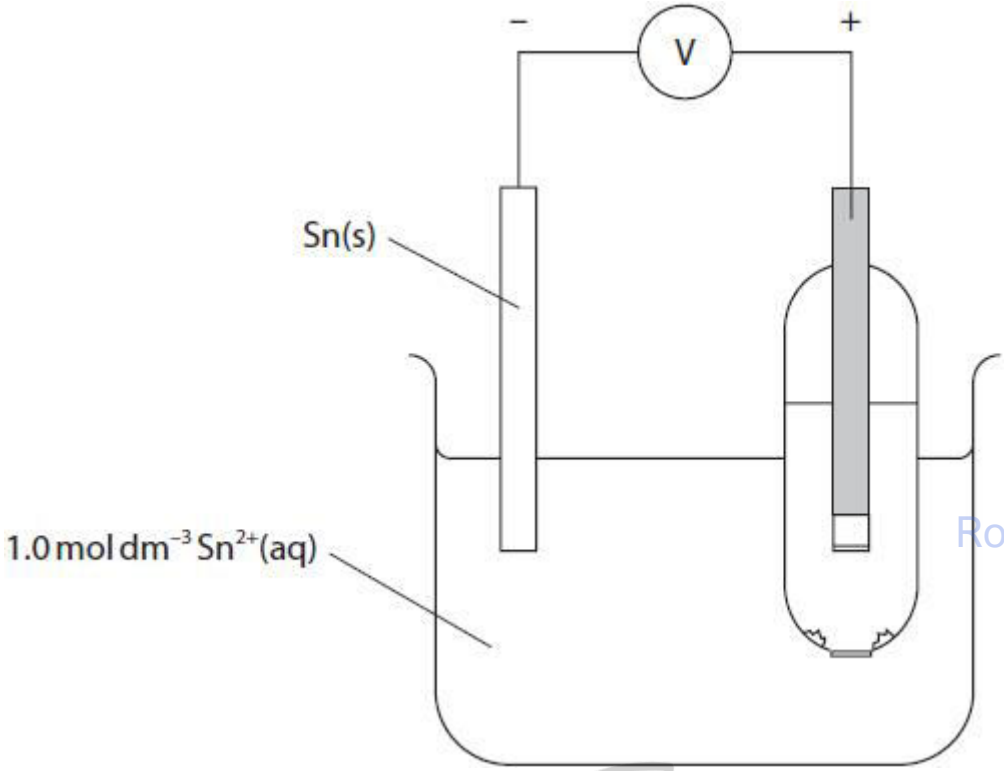
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(ii) A calomel electrode was used to measure the standard electrode potential of the $Sn^{2+}(aq)$ | Sn(s) half-cell.

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The reading on the voltmeter in this cell was +0.37 V.
Deduce the standard electrode potential for the $Sn^{2+}(aq)$ | Sn(s) half−cell.

(1)

(iii) Write the overall equation for the cell reaction.

(1)

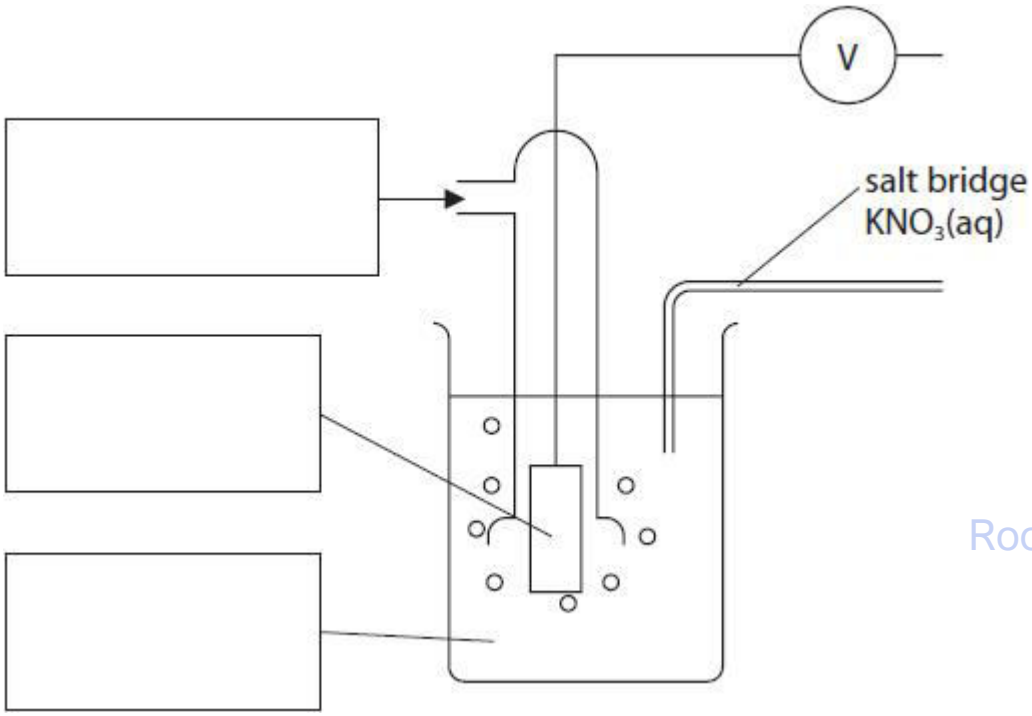
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(iv) Add labels to complete the diagram of a standard hydrogen electrode. Include details of any essential conditions.

(3)

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(v) Suggest **one** advantage of using a calomel electrode, in place of a standard hydrogen electrode, when measuring a cell potential.

(1)

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(Total for question = 20 marks)

Q53.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which complex has a ligand-metal-ligand bond angle of 109.5°?

- ☒ **A** $[CuCl_4]^{2-}$
- ☒ **B** $[Fe(EDTA)]^-$
- ☒ **C** $[Ag(NH_3)_2]^+$
- ☒ **D** $[Pt(NH_3)_2Cl_2]$

(Total for question = 1 mark)

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Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A **small** amount of aqueous ammonia is added to an aqueous solution containing zinc ions. A white precipitate forms.

Which is the equation for this reaction?

☐ **A** $[Zn(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Zn(OH)_2(H_2O)_4]^{2+} + 2NH_4^+$

☐ **B** $[Zn(H_2O)_6]^{2+} + 2NH_3 \rightarrow Zn(OH)_2(H_2O)_4 + 2NH_4^+$

☐ **C** $[Zn(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Zn(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$

☐ **D** $[Zn(H_2O)_6]^{2+} + 4NH_3 \rightarrow Zn(NH_3)_4(H_2O)_2 + 2H_2O + 2H_3O^+$

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(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q55.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which electronic configuration is correct?

☐ **A** Ti^{2+} $[Ar]4s^2$

☐ **B** Cr $[Ar]3d^4 4s^2$

☐ **C** Fe^{3+} $[Ar]3d^3 4s^2$

☐ **D** Cu $[Ar]3d^{10} 4s^1$

(Total for question = 1 mark)

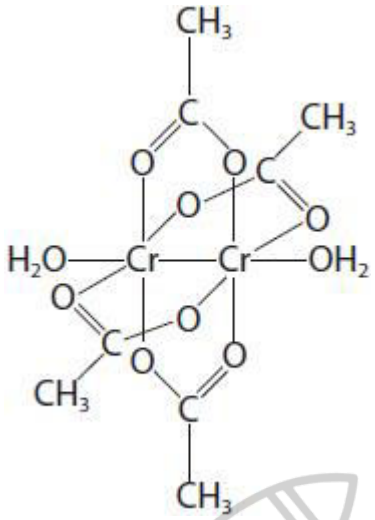
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Q56.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of the complex $Cr_2(CH_3CO_2)_4(H_2O)_2$ is shown.



What is the coordination number of chromium in this complex?

- ☒ **A** two
- ☒ **B** four
- ☒ **C** six
- ☒ **D** twelve

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q57.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the sequence of oxidation number changes for vanadium when V_2O_5 is used in the contact process?

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- ☐ **A** +2 → +1 → +2
- ☐ **B** +2 → +5 → +2
- ☐ **C** +5 → +4 → +5
- ☐ **D** +5 → +6 → +5

(Total for question = 1 mark)

Q58.

This question is about compounds and complex ions of iron.

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(a) A compound of iron contains, by mass, 39.5% potassium, 28.2% iron and 32.3% oxygen.

Calculate the empirical formula of this compound.

(2)

(b) The reaction between iodide ions and persulfate ions is thermodynamically feasible under standard conditions. The reaction is very slow but is catalysed by the presence of Fe^{2+} ions.

$$2I^-(aq) + S_2O_8^{2-}(aq) \rightarrow I_2(s) + 2SO_4^{2-}(aq)$$

(i) Give a reason why the activation energy for this reaction is high in the absence of a catalyst.

(1)

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(ii) Write the ionic equations for the reactions that occur when the catalyst of Fe^{2+} ions is added. State symbols are not required.

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(2)

(c) In aqueous solution, iron(II) exists as the complex ion $[Fe(H_2O)_6]^{2+}$.

(i) Explain how water acts as a monodentate ligand.

(2)

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(ii) Explain the shape of the $[Fe(H_2O)_6]^{2+}$ complex ion.

(2)

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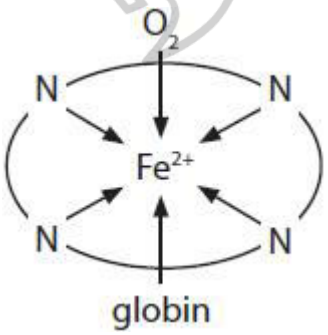
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(d) Haemoglobin is an iron(II) complex which carries oxygen around the body. Part of the structure of oxyhaemoglobin is shown.



The four nitrogen atoms are part of a multidentate ligand in the haem group.
Explain, in terms of the iron(II) complex, why carbon monoxide is toxic.

(2)

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(e) Ethanedioate ligands react with iron(II) ions in aqueous solution.

$$[Fe(H_2O)_6]^{2+} + 2C_2O_4^{2-} \rightarrow [Fe(C_2O_4)_2(H_2O)_2]^{2-} + 4H_2O$$

Explain, in terms of entropy, why this reaction is feasible.

(2)

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(f) Potassium dichromate(VI) oxidises iron(II) ions to iron(III) ions in acid solution.

$$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

Iron(III) ions do not react with acidified potassium dichromate(VI).
A solution **Y** contains 6.28 g dm^{-3} of iron as a mixture of iron(II) ions and iron(III) ions.
50 cm^3 of dilute sulfuric acid is added to a 25.0 cm^3 portion of **Y** and it is titrated with a solution containing 2.56 g dm^{-3} of dichromate(VI) ions, $Cr_2O_7^{2-}$.
The titration is repeated until concordant results are obtained.
The mean titre is 22.55 cm^3.
Calculate the percentage of iron present as **iron(III)** ions in **Y**.

(5)

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(Total for question = 18 marks)

Q59.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A complex ion of chromium(III) has the formula $[CrCl_2(NH_2CH_2CH_2NH_2)_2]^x$, where x is the overall charge.

(a) What is the coordination number of chromium(III) in this complex ion?

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(1)

- **A** 2
- **B** 3
- **C** 4
- **D** 6

(b) What is the overall charge on this complex ion?

(1)

- **A** 0
- **B** +1
- **C** +2
- **D** +3

(Total for question = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q60.

This question is about complex ions.

(a) Which complex ion is square planar?

(1)

- **A** $[Cu(H_2O)_2(NH_3)_4]^{2+}$
- **B** $[CuCl_4]^{2-}$
- **C** $[Pt(NH_3)_2Cl_2]$
- **D** $[Ag(NH_3)_2]^{+}$

(b) Which copper complex ion is colourless?

(1)

- **A** $[CuCl_2]^{-}$
- **B** $[CuCl_4]^{2-}$
- **C** $[Cu(H_2O)_6]^{2+}$
- **D** $[Cu(H_2O)_2(NH_3)_4]^{2+}$

(c) Which complex ion includes a bond angle of 107°?

(1)

- **A** $[Cr(NH_3)_6]^{3+}$
- **B** $[CuCl_2]^{-}$
- **C** $[CuCl_4]^{2-}$
- **D** $[Cr(H_2O)_6]^{2+}$

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(Total for question = 3 marks)

Q61.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which complex ion is colourless?

- ☐ **A** $[Al(H_2O)_6]^{3+}$
- ☐ **B** $[Cu(H_2O)_6]^{2+}$
- ☐ **C** $[Fe(H_2O)_6]^{3+}$
- ☐ **D** $[Ni(H_2O)_6]^{2+}$

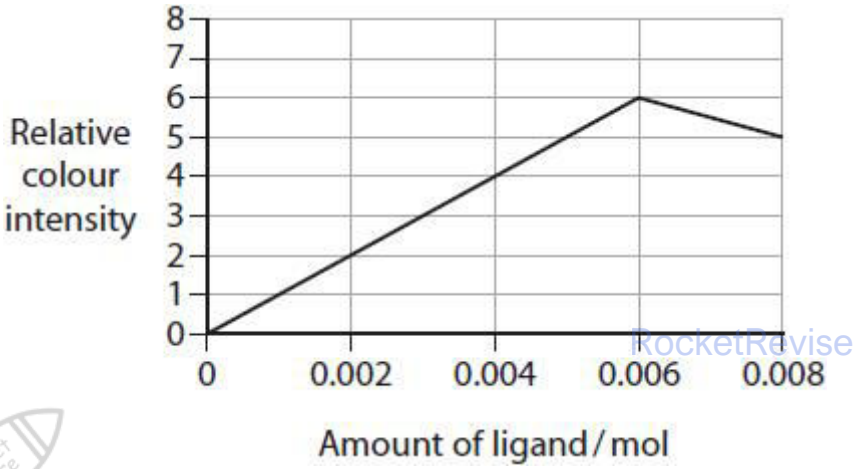
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(Total for question = 1 mark)

Q62.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows how the relative colour intensity of an aqueous solution containing 0.001 mol of chromium(III) ions varies with increasing amounts of cyanide ions, CN^-.



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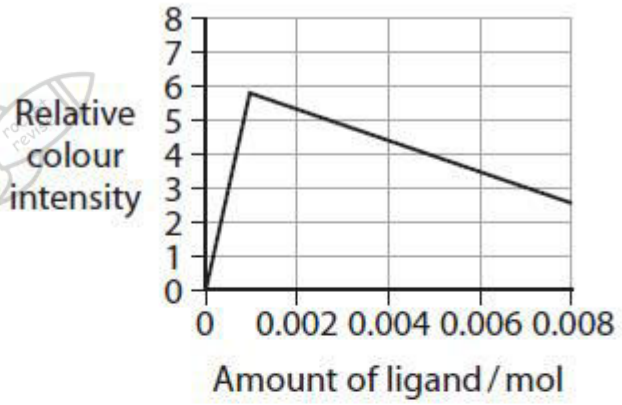
Chromium(III) ions form a complex ion with $EDTA^{4-}$ with a greater relative colour intensity than the complex ion formed with cyanide ions.

Which diagram shows the change in relative colour intensity when increasing amounts of $EDTA^{4-}$ are added to an aqueous solution containing 0.001 mol chromium(III) ions?

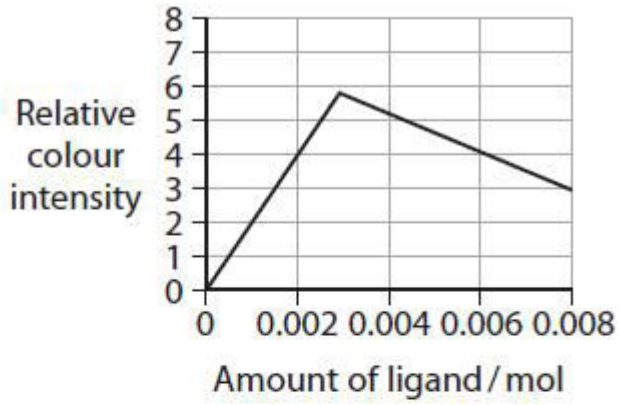
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☒ **A**

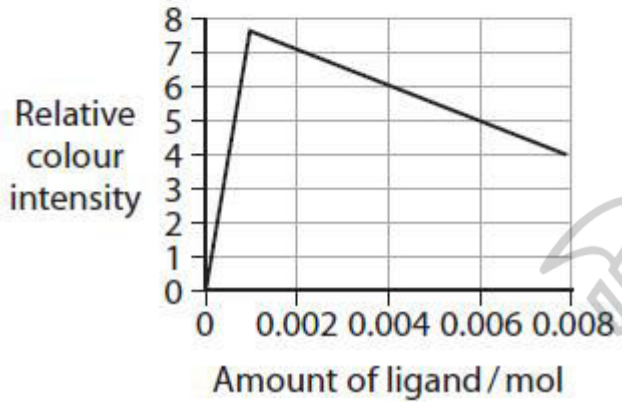


☒ **B**

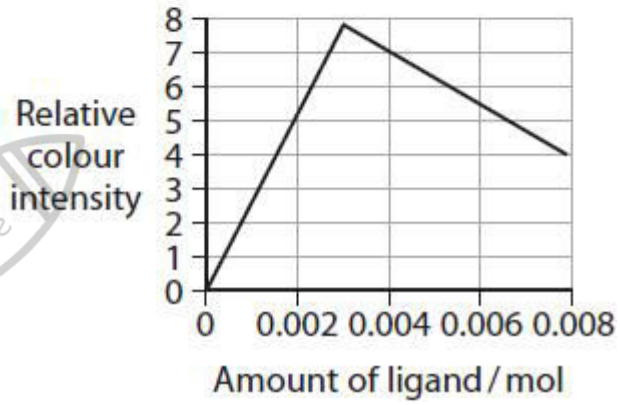


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☒ **C**



☒ **D**



(Total for question = 1 mark)

Q63.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Acidified manganate(VII) ions react with ethanedioate ions.

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$$

	Reason rate increases	Reason rate decreases
☒ **A**	Mn^{2+} acts as a catalyst	kinetic energy of particles decreases
☒ **B**	Mn^{2+} acts as a catalyst	concentration of reactants decreases
☒ **C**	MnO_4^- acts as a catalyst	kinetic energy of particles decreases
☒ **D**	MnO_4^- acts as a catalyst	concentration of reactants decreases

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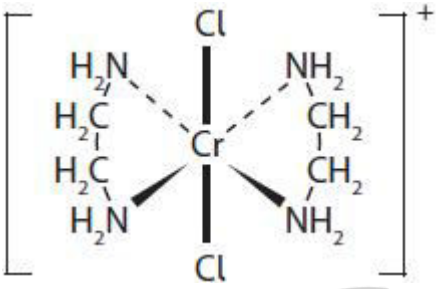
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(Total for question = 1 mark)

Q64.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the co-ordination number of chromium in the complex ion shown?



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- ☐ **A** 2
- ☐ **B** 3
- ☐ **C** 4
- ☐ **D** 6

(Total for question = 1 mark)

Q65.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Aqueous sodium hydroxide and aqueous ammonia are added, until they are in excess, to separate portions of a pale green aqueous solution **E**.

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Test	Observations
Aqueous sodium hydroxide is added to **E**	Green precipitate forms No further change in excess
Aqueous ammonia is added to **E**	Green precipitate forms Green precipitate dissolves in excess to form a pale blue solution

Which ion is present in **E**?

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- ☒ **A** Cu^{2+}
- ☒ **B** Fe^{2+}
- ☒ **C** Ni^{2+}
- ☒ **D** V^{2+}

(Total for question = 1 mark)

Q66.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Aqueous ammonia is added in **excess** to an aqueous solution of Cu^{2+}.

What is the equation for the overall reaction?

- ☒ **A** $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(H_2O)_4(OH)_2] + 2NH_4^+$
- ☒ **B** $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(NH_3)_2(H_2O)_4]^{2+} + 2H_2O$
- ☒ **C** $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$
- ☒ **D** $[Cu(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Cu(NH_3)_6]^{2+} + 6H_2O$

(Total for question = 1 mark)

Q67.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation shows a **redox** reaction in which the solution turns yellow?

- ☒ **A** $Fe + 2HCl \rightarrow FeCl_2 + H_2$
- ☒ **B** $VO_3^- + 2H^+ \rightarrow VO_2^+ + H_2O$
- ☒ **C** $Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$
- ☒ **D** $2Cr(OH)_3 + 3H_2O_2 + 4KOH \rightarrow 2K_2CrO_4 + 8H_2O$

(Total for question = 1 mark)

Q68.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these is a d-block element but is not a transition element?

- ☐ **A** cobalt
- ☐ **B** copper
- ☐ **C** nickel
- ☐ **D** zinc

(Total for question = 1 mark)

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Q69.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation shows the formation of a precipitate?

- ☐ **A** $[Zn(OH)_4]^{2-} + 2H_3O^+ \rightarrow [Zn(H_2O)_4(OH)_2]$
- ☐ **B** $[Zn(H_2O)_6]^{2+} + 4OH^- \rightarrow [Zn(OH)_4]^{2-} + 6H_2O$
- ☐ **C** $[Cr(H_2O)_3(OH)_3] + 3H_3O^+ \rightarrow [Cr(H_2O)_6]^{3+} + 3H_2O$
- ☐ **D** $[Cr(H_2O)_3(OH)_3] + 3OH^- \rightarrow [Cr(OH)_6]^{3-} + 3H_2O$

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q70.

Brass is a metal alloy containing copper and zinc. The presence of zinc in the alloy makes brass less malleable than copper alone.
Prince's metal is one type of brass. It is used to make imitation gold because of its yellow colour.

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The copper content of brass can be analysed by first reacting a known sample of the metal with concentrated nitric acid. The reaction of the copper is:

$$Cu + 4H^+ + 2NO_3^- \rightarrow Cu^{2+} + NO_2 + 2H_2O.$$

(a) Identify the element that is oxidised and the element that is reduced in the reaction shown. Include relevant oxidation numbers.

(2)

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(b) Suggest one precaution when carrying out this reaction, other than the use of gloves, goggles and lab coats, clearly stating the hazard concerned.

(2)

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(c) The copper ions are then reacted with excess potassium iodide.

$$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$$

The iodine formed is analysed by titration with sodium thiosulfate.

$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

A 5.000 g sample of Prince's metal was analysed.
After reaction with concentrated nitric acid, the sample was diluted to 250 cm^3 and then 10.0 cm^3 aliquots or portions were titrated with 0.100 mol dm^{-3} sodium thiosulfate solution.
The mean titre was 22.65 cm^3.
Calculate the percentage of copper, by mass, in this sample of Prince's metal to an appropriate number of significant figures.

(6)

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(d) In aqueous solution, copper(II) and zinc ions react differently with sodium hydroxide solution.

Describe the observations when sodium hydroxide solution is added drop by drop (until in excess) to separate samples of these two ions.
Include relevant **ionic** equations with state symbols.

(6)

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(e) Explain, in terms of electronic configurations, why copper is classified as a transition element but zinc is not.

(2)

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(f) Explain, in terms of their structures, why brass is less malleable than pure copper.

(2)

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(Total for question = 20 marks)

Q71.

Which vanadium ion is yellow in aqueous solution?

(1)

- ☒ **A** VO^{2+}
- ☒ **B** VO^{2} +
- ☒ **C** V^{3+}
- ☒ **D** V^{2+}

(Total for question = 1 mark)

Q72.

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This question is about silver and silver compounds.

Glass decorations are made reflective by coating their inner surface with silver. This is achieved by using the reaction between silver nitrate solution, ammonia and glucose, under alkaline conditions.

Initially the colourless complex ion diamminesilver(I), $[Ag(NH_3)_2]^+$, forms.

(a) (i) Explain the shape of $[Ag(NH_3)_2]^+$.

(3)

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(ii) Explain why $[Ag(NH_3)_2]^+$ is colourless.

(2)

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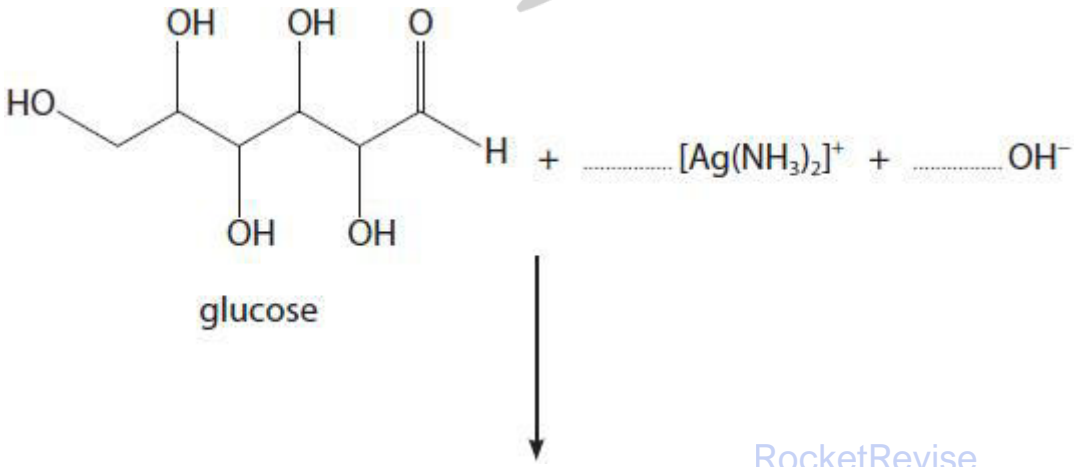
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(b) The diamminesilver(I) complex then reacts with glucose forming silver and an organic compound, **Y**. Two other products also form.

(i) Complete the equation for the reaction.

(2)



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$C_6H_{12}O_7$ + ...Ag + ... + ...

(ii) Draw the structure of **Y**.

(1)

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(c) The overall reaction in a silver cell used in watch batteries is

$$Ag_2O(s) + H_2O(l) + Zn(s) \rightarrow 2Ag(s) + Zn(OH)_2(s)$$

The half-equation for the reaction at the positive electrode is

$$Ag_2O(s) + H_2O(l) + 2e^- \rightarrow 2Ag(s) + 2OH^-(aq)$$

Deduce the half-equation for the reaction at the negative electrode.
State symbols are **not** required.

(1)

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(Total for question = 9 marks)

Q73.

Which reagent will be **most** effective at shifting the equilibrium towards the chromate(VI) ions?

$$Cr_2O_7^{2-}(aq) + H_2O(l) \rightleftharpoons 2CrO_4^{2-}(aq) + 2H^+(aq)$$

(1)

- ☒ **A** hydrochloric acid
- ☒ **B** sulfuric acid
- ☒ **C** sodium hydroxide
- ☒ **D** water

(Total for question = 1 mark)

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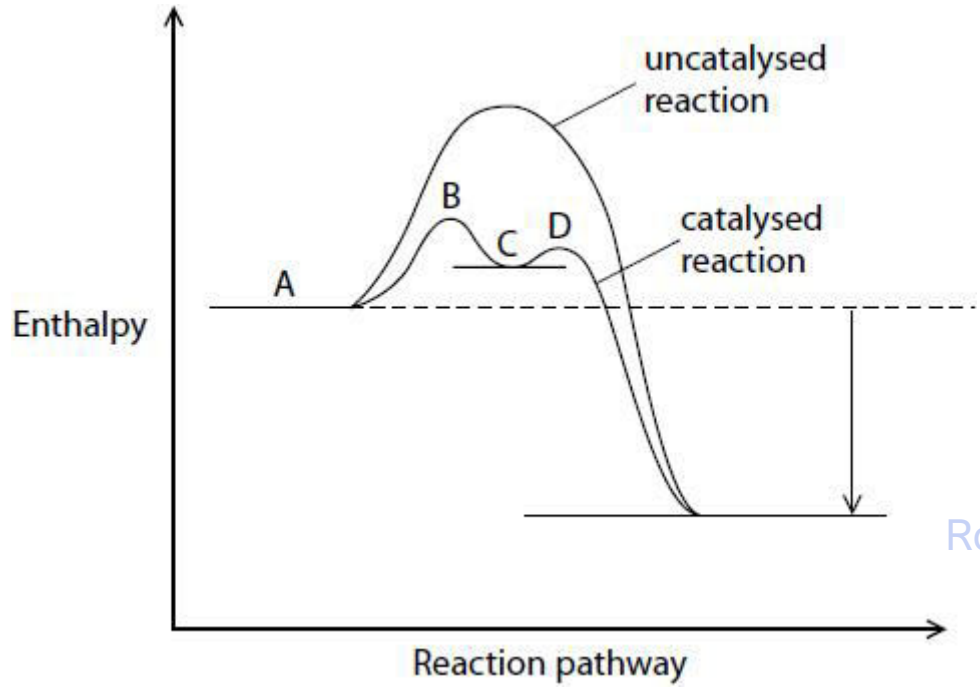
Q74.

In homogeneous catalysis, the catalyst is in the same state as the reactants.

(a) The enthalpy profile diagram for a homogeneously catalysed reaction is

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Which label indicates the intermediate species?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(b) Iodide ions can be oxidised by peroxodisulfate(VI) ions in the reaction shown.

$$2I^-(aq) + S_2O_8^{2-}(aq) \rightarrow I_2(aq) + 2SO_4^{2-}(aq)$$

What property of iron(III) ions enables them to act as homogeneous catalysts for this reaction?

(1)

- ☒ **A** they can be oxidised and then reduced
- ☒ **B** they can gain and then lose electrons
- ☒ **C** they provide an effective surface for reaction to occur on
- ☒ **D** they can form complex ion intermediates with a lower activation energy

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(Total for question = 2 marks)

Q75.

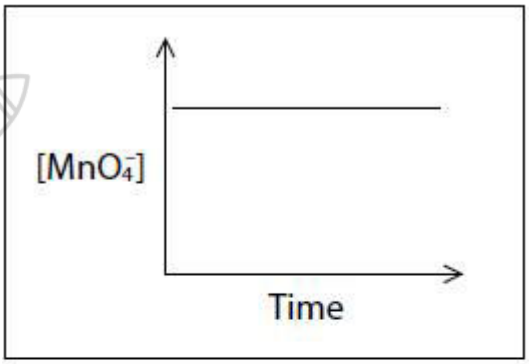
Which sketch shows the change in concentration of manganate(VII) ions with time in the reaction?

$$2MnO_4^-(aq) + 16H^+(aq) + 5C_2O_4^{2-}(aq) \rightarrow 10CO_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$$

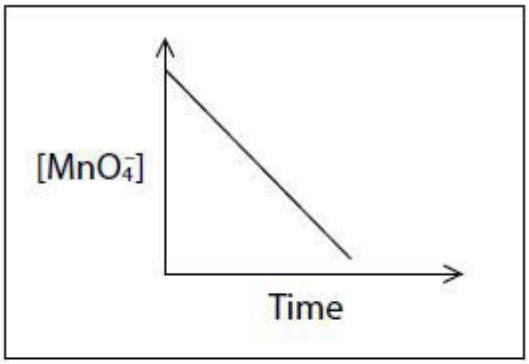
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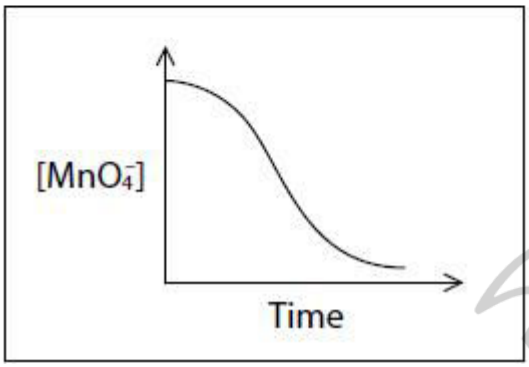
☐ **A**



☐ **B**

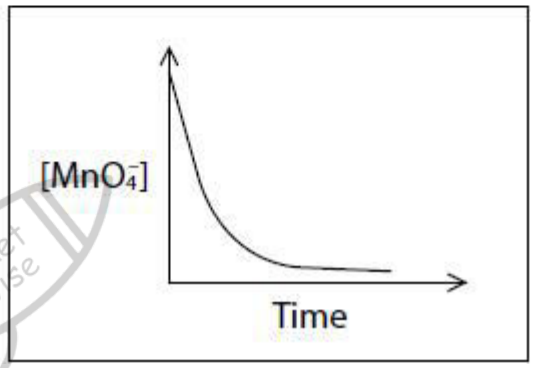


☐ **C**



☐ **D**

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(Total for question = 1 mark)

Q76.

Which is **not** a bidentate ligand?

(1)

- ☐ **A** ethanedioate ion, $C_2O_4^{2-}$
- ☐ **B** ethanoate ion, CH_3COO^-
- ☐ **C** 1,2-diaminoethane, $NH_2CH_2CH_2NH_2$
- ☐ **D** 2-aminoethanoic acid, NH_2CH_2COOH

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(Total for question = 1 mark)

Q77.

* Compare and contrast the reactions in aqueous solution of nickel(II) sulfate with sodium hydroxide and with ammonia.

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For each reaction include

- what would be seen
- the equation (state symbols are not required)
- the type of reaction.

(Total for question = 6 marks)

Q78.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

An oxide of vanadium is the catalyst in one step of the contact process for the manufacture of sulfuric acid.

Which reaction involves the vanadium oxide catalyst and how does the oxidation number of vanadium change during the process?

		Reaction	Oxidation number changes for vanadium
☐	**A**	$S + O_2 \rightarrow SO_2$	$+5 \rightarrow +4 \rightarrow +5$
☐	**B**	$SO_2 \rightarrow SO_3$	$+4 \rightarrow +5 \rightarrow +4$
☐	**C**	$SO_2 \rightarrow SO_3$	$+5 \rightarrow +4 \rightarrow +5$
☐	**D**	$SO_3 + H_2O \rightarrow H_2SO_4$	$+4 \rightarrow +5 \rightarrow +4$

(Total for question = 1 mark)

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Q79.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Car exhaust gases react on the surface of a catalytic converter.

$$2CO + 2NO \rightarrow 2CO_2 + N_2$$

Which interactions could **not** occur between the gases and the catalytic converter in this

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process?

- ☒ **A** covalent bonds
- ☒ **B** hydrogen bonds
- ☒ **C** London forces
- ☒ **D** permanent dipole-induced dipole forces

(Total for question = 1 mark)

Q80.

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This question is about cell reactions involving chromium.

Use your Data Booklet when answering this question.

(a) (i) Name the type of the forward reaction that is shown in the right-hand electrode systems in the Data Booklet.

(1)

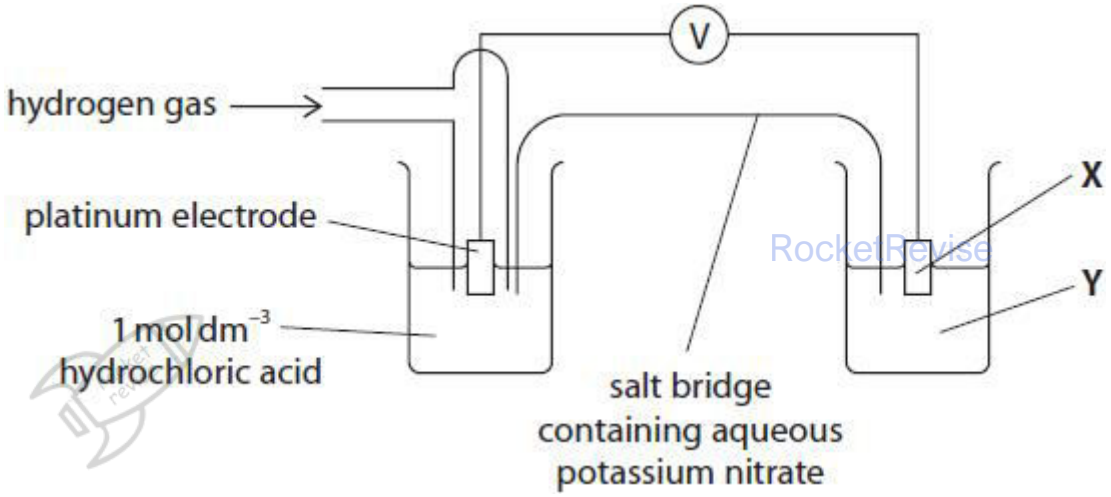
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(ii) Name the series formed when the right-hand electrode systems are placed in order, most negative first.

(1)

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(b) A student set up the following apparatus to measure the standard electrode potential for right-hand electrode system 8 in your Data Booklet.



(i) Identify **X** and **Y**.

(2)

X: .....

Y: .....

(ii) Give **two** reasons why the initial reading on the voltmeter may differ from the stated value

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given in the Data Booklet.

(2)

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(iii) The voltmeter is removed and the cell is allowed to run for one hour.
Explain the changes that would occur in the right-hand half-cell during this time.

(2)

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(c) (i) Explain, by calculating $E^{\ominus}_{cell}$ values, why Fe(II) is used to reduce Cr(VI) to Cr(III) but zinc is used to reduce Cr(VI) to Cr(II).

(4)

Numbers chosen of the right-hand electrode systems from the Data Booklet

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Explanation

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(ii) State the essential condition required for these reactions to occur.

(1)

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(Total for question = 13 marks)

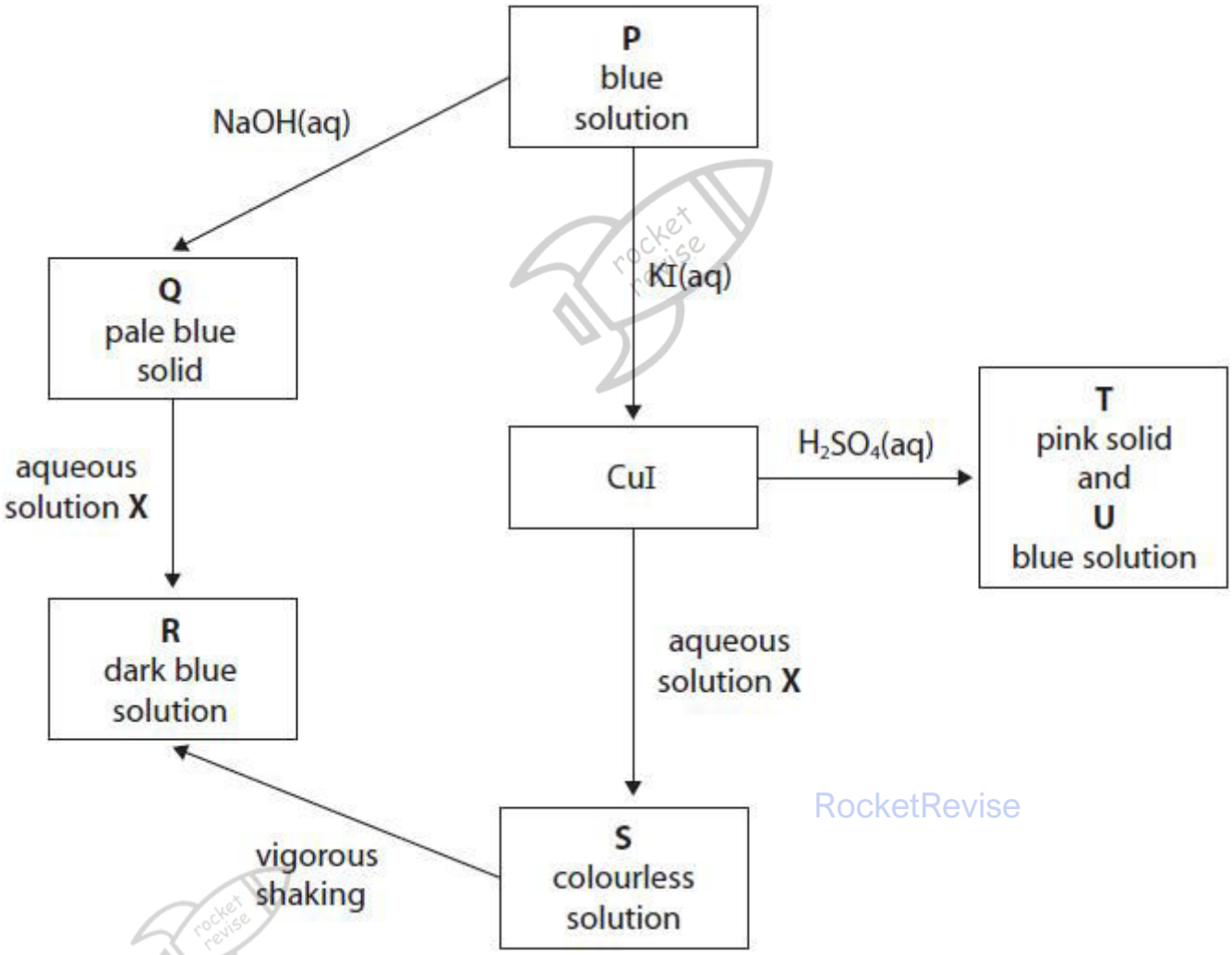
Q81.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question concerns the chemistry of copper.

In the reaction sequence, the substances P, Q, R, S, T and U contain copper in various oxidation states.

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(a) What are the electronic structures of the three copper species?

(1)

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	Cu	Cu^+	Cu^{2+}
☐ **A**	[Ar] $3d^9\ 4s^2$	[Ar] $3d^9\ 4s^1$	[Ar] $3d^9$
☐ **B**	[Ar] $3d^9\ 4s^2$	[Ar] $3d^8\ 4s^2$	[Ar] $3d^7\ 4s^2$
☐ **C**	[Ar] $3d^{10}\ 4s^1$	[Ar] $3d^9\ 4s^1$	[Ar] $3d^8\ 4s^1$
☐ **D**	[Ar] $3d^{10}\ 4s^1$	[Ar] $3d^{10}$	[Ar] $3d^9$

(b) Which rows show the substances with their correct oxidation states?

(1)

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	Cu(0)	Cu(I)	Cu(II)
☐ **A**	S	U	P
☐ **B**	S	R	Q
☐ **C**	T	S	P
☐ **D**	T	U	Q

(c) Aqueous solution **X** contains

(1)

- ☐ **A** H_2SO_4
- ☐ **B** KI
- ☐ **C** NaOH
- ☐ **D** NH_3

(Total for question = 3 marks)

Q82.

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This question is about coloured transition metal complex ions.

(a) The $[Fe(H_2O)_6]^{2+}$ complex ion is green and the $[Fe(H_2O)_6]^{3+}$ complex ion is yellow.

(i) Explain why the $[Fe(H_2O)_6]^{2+}$ complex ion is coloured.

(3)

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(ii) Explain why the two iron complex ions are different colours.

(2)

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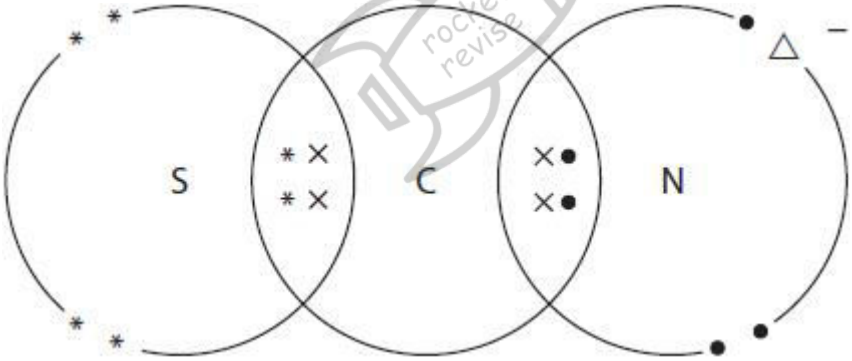
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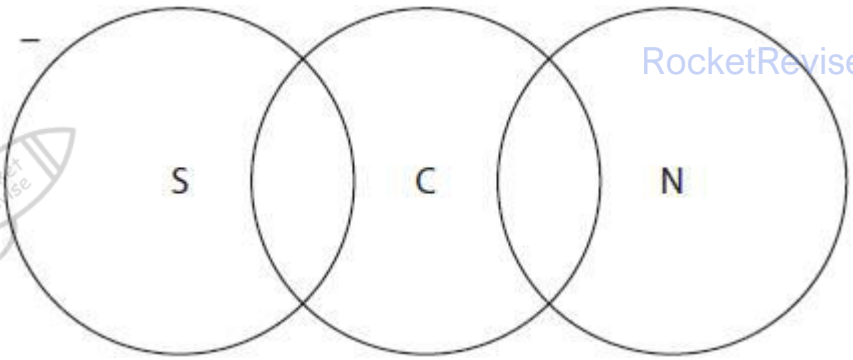
(b) The thiocyanate ion, SCN^-, forms a 'blood-red' coloured complex ion with iron(III) ions.

(i) The dot-and-cross diagram for the thiocyanate ion can be drawn in more than one way. For example, the negative charge can be on the nitrogen atom as shown.



Complete the alternative dot-and-cross diagram of the thiocyanate ion where the negative charge is on the sulfur atom.
Use (•) for the nitrogen electrons, (×) for the carbon electrons, (*) for the sulfur electrons and (Δ) for the extra electron.

(2)



(ii) 12.8 cm^3 of 0.05 mol dm^{-3} iron(III) chloride solution reacted with 8.0 cm^3 of 0.08 mol dm^{-3} ammonium thiocyanate to form a complex ion.
Deduce the formula of the **octahedral** complex ion that iron(III) ions form with thiocyanate ions in aqueous solution.
You **must** show your working.

(3)

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Formula

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(c) Colorimetry was used to determine the concentration of a solution of copper(II) sulfate.

Six solutions of copper(II) sulfate of known concentration were prepared and their absorbance at a wavelength of 635 nm measured. The results are shown.

Concentration of $CuSO_4(aq)$ / mol dm^{-3}	0.00	0.10	0.20	0.30	0.40	0.50
Absorbance at 635 nm	0.00	0.28	0.55	0.83	1.10	1.38

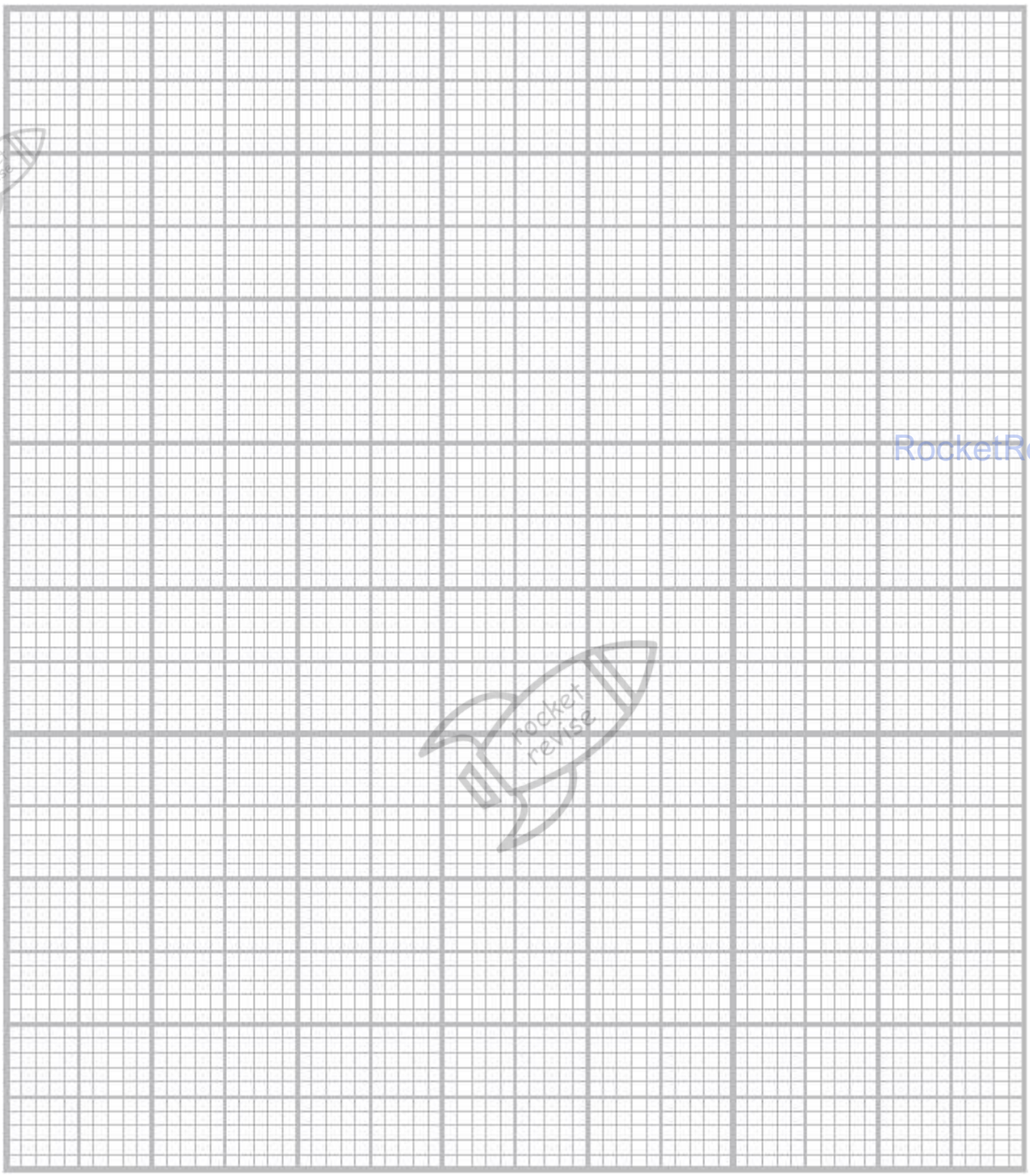
(i) Plot a graph of absorbance against concentration using the data in the table.

(3)

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(ii) 50.0 cm^3 of a solution of copper(II) sulfate of unknown concentration was pipetted into a 250.0 cm^3 volumetric flask and the volume made up to the mark with deionised water.

The absorbance at a wavelength of 635 nm for this diluted copper(II) sulfate solution was 0.72.

Determine the concentration of the **original** copper(II) sulfate solution.

You **must** show your working on the graph.

(2)

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(iii) Give **one** possible reason why any concentrations of copper(II) sulfate determined from the graph of values over 0.50 mol dm^{-3} are uncertain.

(1)

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(d) The complex ion formed from cobalt(III) ions and ethane-1,2-diamine is a yellow-orange colour.

The equation for the formation of this complex ion is shown.

$$[Co(H_2O)_6]^{3+} + 3H_2NCH_2CH_2NH_2 \rightarrow [Co(H_2NCH_2CH_2NH_2)_3]^{3+} + 6H_2O$$

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(i) Give a reason, by referring to the equation, why the formation of the diamine complex from the aqueous complex ion is thermodynamically favoured.

(1)

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(ii) Explain, by reference to the structure of ethane-1,2-diamine, why it is a bidentate ligand.

(2)

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(iii) Explain how the H—N—H bond angles would be expected to change in ethane-1,2-diamine when the complex ion is formed.

(3)

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(Total for question = 22 marks)

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Q83.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

What colour is vanadium(III) in aqueous solution?

- ☐ **A** blue
- ☐ **B** green
- ☐ **C** purple
- ☐ **D** yellow

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(Total for question = 1 mark)

Q84.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which pair of equations shows the most likely mechanism for vanadium(V) oxide acting as a catalyst in the contact process?

- ☐ **A** $V_2O_5 + \frac{1}{2}O_2 \rightarrow V_2O_6$ then $V_2O_6 + SO_2 \rightarrow V_2O_5 + SO_3$
- ☐ **B** $V_2O_5 + SO_2 \rightarrow V_2O + 1\frac{1}{2}O_2 + SO_3$ then $V_2O + 2O_2 \rightarrow V_2O_5$
- ☐ **C** $V_2O_5 + \frac{1}{2}O_2 \rightarrow 2VO_3$ then $2VO_3 + SO_2 \rightarrow V_2O_5 + SO_3$
- ☐ **D** $V_2O_5 + SO_2 \rightarrow V_2O_4 + SO_3$ then $V_2O_4 + \frac{1}{2}O_2 \rightarrow V_2O_5$

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(Total for question = 1 mark)

Q85.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The formation of carbon dioxide from the reaction between manganate(VII) ions and

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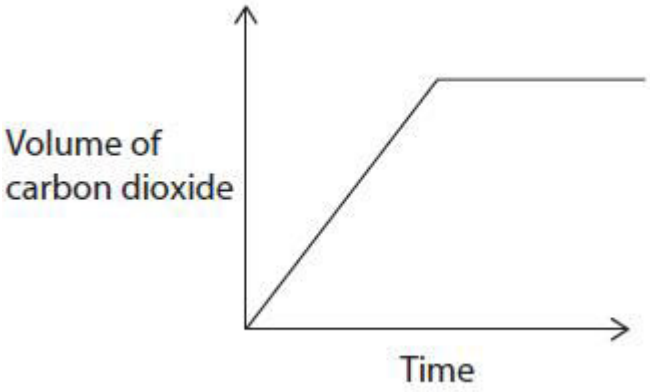
ethanedioate ions is autocatalysed by manganese(II) ions.

The equation for this reaction is shown.

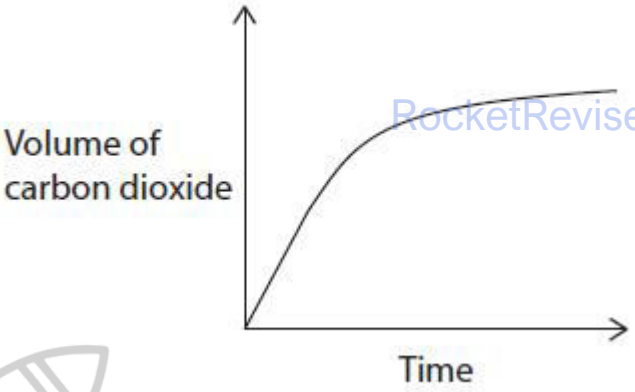
$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$$

Which graph for the complete reaction shows the autocatalysis of this reaction by the manganese(II) ions?

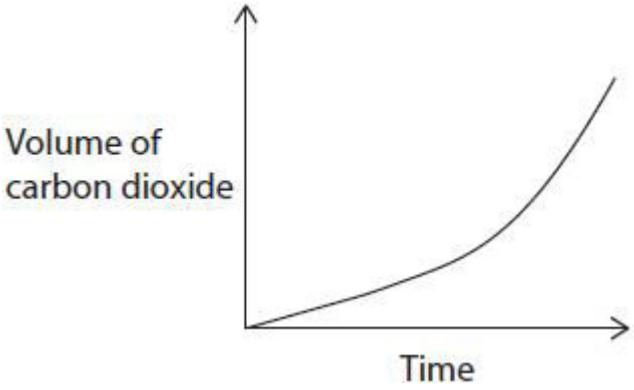
☐ **A**



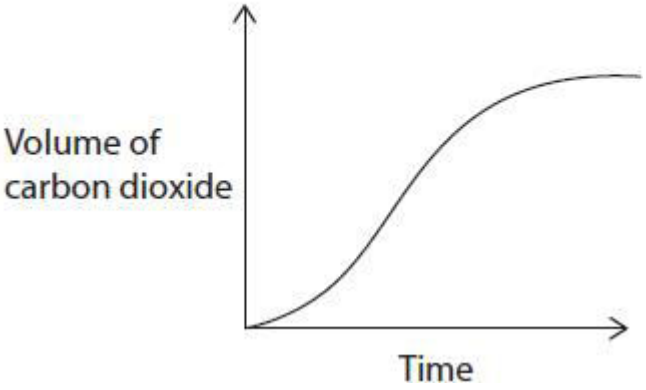
☐ **B**



☐ **C**



☐ **D**



(Total for question = 1 mark)

Q86.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Chromium(III) hydroxide, $Cr(OH)_3(H_2O)_3$, is amphoteric.

This means that chromium(III) hydroxide dissolves in

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- ☒ **A** water
- ☒ **B** strong acid only
- ☒ **C** strong alkali only
- ☒ **D** strong acid and strong alkali

(Total for question = 1 mark)

Q87.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which successive ionisation energies (in kJ mol^{-1}) are most likely to be those of a transition element?

- ☒ **A** 578 1817 2745 11578 14831
- ☒ **B** 759 1561 2958 5290 7236
- ☒ **C** 789 1577 3232 4356 16091
- ☒ **D** 801 2427 3660 25026 32828

(Total for question = 1 mark)

Q88.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

What is the electronic configuration of the chromium **atom**?

- ☒ **A** [Ar] $3d^4\ 4s^2$
- ☒ **B** [Ar] $3d^5\ 4s^1$
- ☒ **C** [Ar] $3d^6$
- ☒ **D** [Ar] $3d^1\ 4s^2\ 4p^3$

(Total for question = 1 mark)

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Q89.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The reaction between peroxodisulfate ions and iodide ions is shown.

$$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$$

The reaction has a high activation energy but is catalysed by iron(II) ions.

(a) What is the best explanation for the high activation energy of this reaction?

(1)

- ☐ **A** the peroxodisulfate ion is a weak oxidising agent
- ☐ **B** the iodide ion is a weak reducing agent
- ☐ **C** peroxodisulfate ions and iodide ions are both negatively charged
- ☐ **D** the reaction involves three reactant ions

(b) Which property of iron(II) ions enables them to function as a catalyst for this reaction?

(1)

- ☐ **A** iron(II) ions can be easily reduced to metallic iron
- ☐ **B** iron(II) ions effectively adsorb the reactants
- ☐ **C** iron(II) ions can be easily oxidised and then reduced
- ☐ **D** iron(II) ions can be easily reduced and then oxidised

(Total for question = 2 marks)

Q90.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which ion shows the typical property of transition metals by forming coloured complexes?

- ☐ **A** Cu^+
- ☐ **B** Ni^{2+}
- ☐ **C** Sc^{3+}
- ☐ **D** Zn^{2+}

(Total for question = 1 mark)

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Q91.

Transition metal compounds can show a number of different types of isomerism. Hydration isomerism is where different numbers of water molecules act as ligands. The name chromium(III) chloride is given to several chemical compounds with the formula $CrCl_3 \cdot xH_2O$, including a number of hydration isomers.

(a) Anhydrous chromium(III) chloride, $CrCl_3$, is a violet solid which can react with water to produce three isomers.

$[Cr(H_2O)_6]^{3+}3Cl^-$ is violet.

$[Cr(H_2O)_5Cl]^{2+}2Cl^- \cdot H_2O$ is pale green.

$[Cr(H_2O)_4Cl_2]^+Cl^- \cdot 2H_2O$ is dark green.

(i) Explain why the three isomers have different colours.

(3)

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(ii) You are provided with three equimolar solutions of the three isomers. Suggest how you would quantitatively determine the relative numbers of free chloride ions in the three isomers using the standard test for a chloride ion.

(5)

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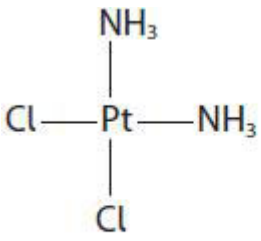
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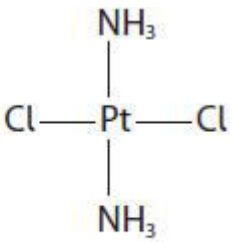
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(b) Complexes can exhibit both cis-trans and optical isomerism. The *cis*- and *trans*-isomers of diamminedichloroplatinum(II) are commonly known as *cis*-platin and *trans*-platin.



cis-platin



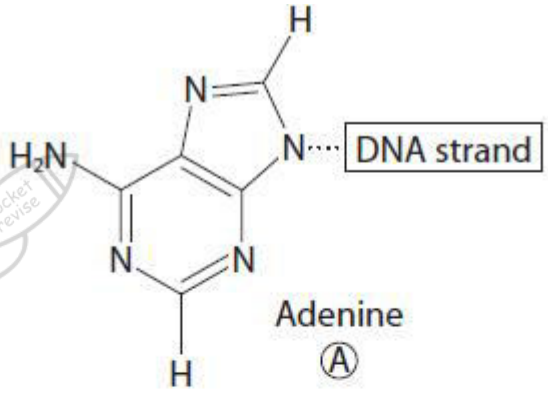
trans-platin

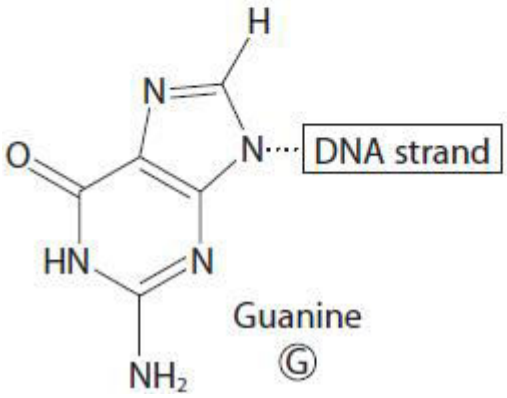
In chemotherapy medication, *cis*-platin is used to treat a number of cancers including testicular cancer and breast cancer, while *trans*-platin has no beneficial effect against cancer. The cis-isomer is effective because it binds with the deoxyribonucleic acid (DNA) molecules in a cancerous cell through adenine and guanine groups. This interferes with the replication of the cell and results in its destruction.

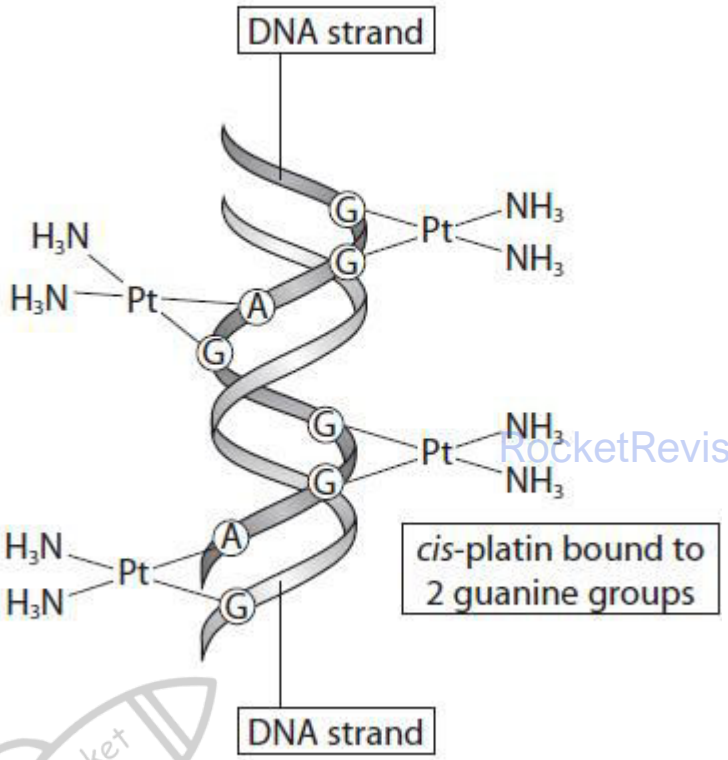
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This works in three steps.
Step **1** Slow substitution of one chloride ligand by a water.
Step **2** This water ligand is easily displaced by guanine or adenine in the DNA strand.
Step **3** Finally the second chloride ligand is displaced by a different guanine or adenine from a different part of the strand.

(i) Write the balanced equation for Step **1**.
State symbols are not required.

(1)

(ii) Describe how guanine and adenine can bind to the platinum ion.

(1)

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(iii) *trans*-Platin binds to DNA.
Suggest why this does not damage the DNA.

(2)

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(c) Reinecke's salt has the formula $NH_4[Cr(SCN)_x(NH_3)_y] \cdot zH_2O$. The anion exists as a trans–isomer.

It contains 14.67% chromium, 36.23% sulfur, 4.51% oxygen and 27.65% nitrogen by mass.

(i) Calculate the values x, y and z in Reinecke's salt.
You **must** show your working.

(3)

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(ii) Draw a diagram of the **anion** of Reinecke's salt showing its three-dimensional shape.

(2)

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(d) (i) Explain why the tetrahedral complex $[Co(NH_3)ClBrI]$ exists as two optical isomers.

(2)

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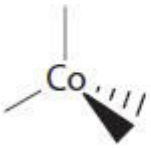
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(ii) Complete the diagram showing the **two** optical isomers of the tetrahedral complex.

(1)



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(Total for question = 20 marks)

Q92.

Metal ions occur naturally in river water but near industrial plants the concentrations can reach toxic levels. Wastewater from a stainless steel electropolishing plant contains dangerous concentrations of transition metal ions as aqueous complexes.

Chemical precipitation can be used to remove these ions from the water.

(a) Complete the equation for the reaction of a laboratory reagent with aqueous Fe^{3+} ions to produce a precipitate.

Include state symbols in your answer.

(2)

$[Fe(H_2O)_6]^{3+}(aq)$ + $\rightarrow$

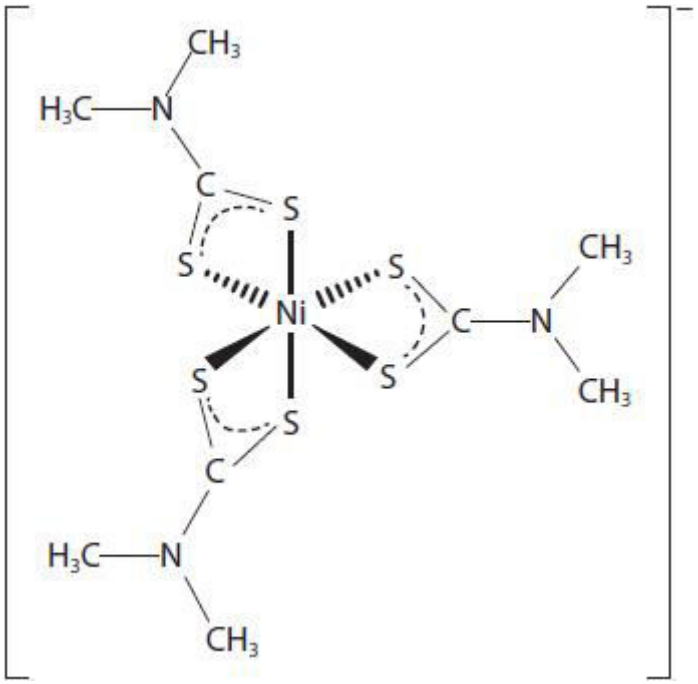
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(b) Sodium dimethyldithiocarbamate, $(CH_3)_2NCS_2^-Na^+$, can also be used to precipitate metal ions from wastewater in acidic solutions.

A representation of the molecule bonding with a nickel ion is shown.

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(i) Complete the table for the complex.

(3)

Charge on nickel ion	
Type of ligand	
Co-ordination number of the nickel ion	
Shape of metal complex around the nickel ion	

(ii) Explain, using a balanced equation, why the formation of the dimethyldithiocarbamate complex from an aqueous solution of nickel ions is thermodynamically feasible.
State symbols are not required.

(2)

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(iii) When a 5 dm^3 sample of wastewater was treated with excess **acidified** sodium dimethyldithiocarbamate, 245.0 mg of the dry precipitate $H^+[Ni((CH_3)_2NCS_2)_3]^-$ was formed.
Calculate the concentration, in mol dm^{-3}, of nickel ions in the sample.
[Molar mass of dimethyldithiocarbamate ion = 120.2 g mol^{-1}]

(3)

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(Total for question = 10 marks)

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Q93.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these pairs of elements are both transition metals?

- ☐ **A** gold and scandium
- ☐ **B** gold and titanium
- ☐ **C** scandium and zinc
- ☐ **D** titanium and zinc

(Total for question = 1 mark)

Q94.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What are the shapes of the complexes $[CuCl_4]^{2-}$ and $Pt(NH_3)_2Cl_2$?

	$[CuCl_4]^{2-}$	$Pt(NH_3)_2Cl_2$
☐ **A**	square planar	square planar
☐ **B**	square planar	tetrahedral
☐ **C**	tetrahedral	square planar
☐ **D**	tetrahedral	tetrahedral

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(Total for question = 1 mark)

Q95.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which species **cannot** act as a ligand in the formation of a complex ion?

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- ☒ **A** C_2H_6
- ☒ **B** CO
- ☒ **C** OH^-
- ☒ **D** $C_4H_9NH_2$

(Total for question = 1 mark)

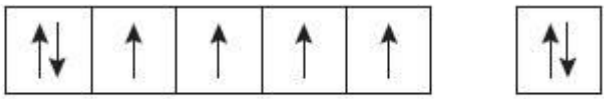
Q96.

This question is about iron(II) sulfate, $FeSO_4$.

(a) (i) The electronic configuration for Fe is shown.

(1)

Fe [Ar]

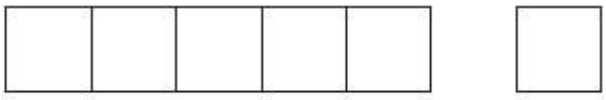


3d 4s

Complete the electronic configuration for the Fe^{2+} ion in iron(II) sulfate.

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Fe^{2+} [Ar]



3d 4s

(ii) An aqueous solution of iron(II) sulfate is pale green in colour. When left exposed to the air the solution turns brown.

Explain why this colour change occurs. In your response refer to your answer from (a)(i).

(3)

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(b) Many moss killers contain iron(II) sulfate, $FeSO_4$, as the active compound. Potassium manganate(VII) can be used to determine the amount of iron(II) sulfate present.

In an experiment, 6.42 g of moss killer was transferred to a volumetric flask and made up to 250 cm^3 with distilled water. A 25.0 cm^3 portion of this solution was acidified and titrated with 0.00740 mol dm^{-3} potassium manganate(VII) solution.

The equation for the titration is shown.

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$$

The mean titre was 17.70 cm^3.

Calculate the percentage, by mass, of iron(II) sulfate in the moss killer.

Give your answer to an appropriate number of significant figures.

[Molar mass $FeSO_4$ = 151.9 g mol^{-1}]

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(6)

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(c) Suggest why iron(II) sulfate is acidic when dissolved in water.

(2)

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(Total for question = 12 marks)

Q97.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The first five ionisation energies of four elements are shown.

Which are the ionisation energies of a transition metal?

☐ **A** 419 3051 4412 5877 7975

☐ **B** 578 1817 2745 11578 14831

☐ **C** 738 1451 7733 10541 13629

☐ **D** 758 1646 3232 4950 7671

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(Total for question = 1 mark)

Q98.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An aqueous solution of a transition metal ion absorbs frequencies of light corresponding to

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green, yellow and red.

Which ion is likely to be in the solution?

- ☒ **A** Mn^{2+}
- ☒ **B** Co^{2+}
- ☒ **C** Ni^{2+}
- ☒ **D** Cu^{2+}

(Total for question = 1 mark)

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Q99.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the colour of the VO^{2+} ion in aqueous solution?

- ☒ **A** blue
- ☒ **B** green
- ☒ **C** violet
- ☒ **D** yellow

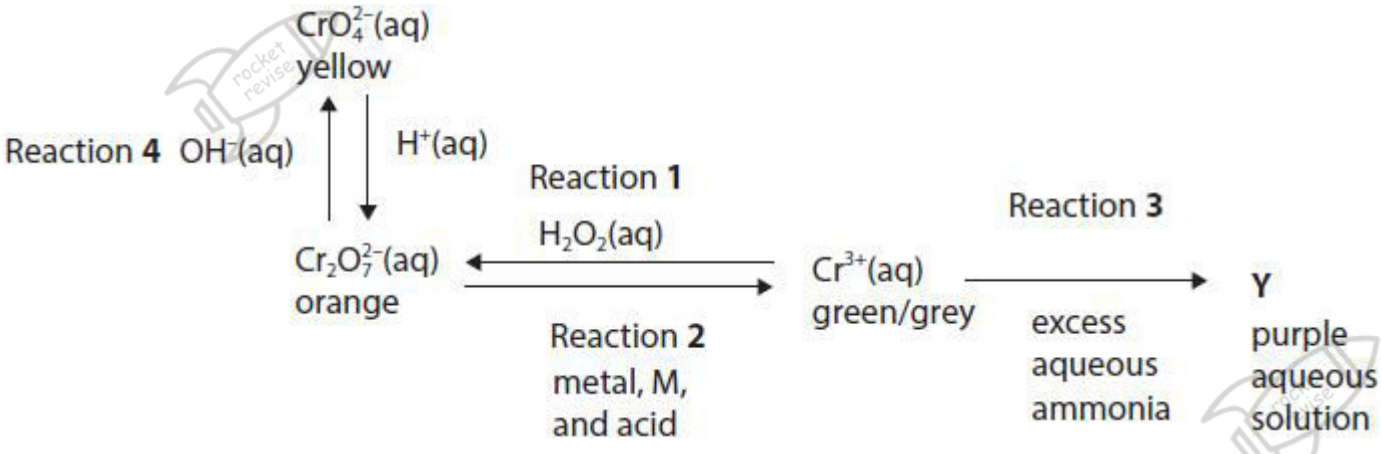
(Total for question = 1 mark)

Q100.

This question is about some chromium compounds.

(a) Some reactions of chromium compounds are shown.

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(i) Write the equation for Reaction **1**. Use the half-equations from page 10 of your Data Booklet. State symbols are **not** required.

(2)

(ii) In Reaction **2**, $Cr_2O_7^{2-}$ reacts with a metal, M, in acid conditions forming Cr^{3+} and M^{2+}.
Identify a possible metal, M, calculating $E^{\ominus}_{cell}$ for its reaction with $Cr_2O_7^{2-}$.
Use information from page 10 of your Data Booklet.

(2)

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(iii) Give the formula of the species responsible for the purple colour formed in Reaction **3**, naming the type of reaction that occurs.

(2)

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(iv) State whether or not the reaction in Reaction **4** is a redox reaction. Justify your answer using oxidation numbers.

(2)

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(b) A double salt of chromium contains Cr^{3+}, K^+ and SO_4^{2-} ions.

(i) Deduce the simplest formula of this double salt.

(1)

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(ii) The double salt also exists as hydrated crystals.
When a sample of the hydrated double salt is heated until all the water of crystallisation is removed, 56.74 % of the original mass of solid remains.
Calculate the number of moles of water of crystallisation in one mole of the hydrated double salt.

(4)

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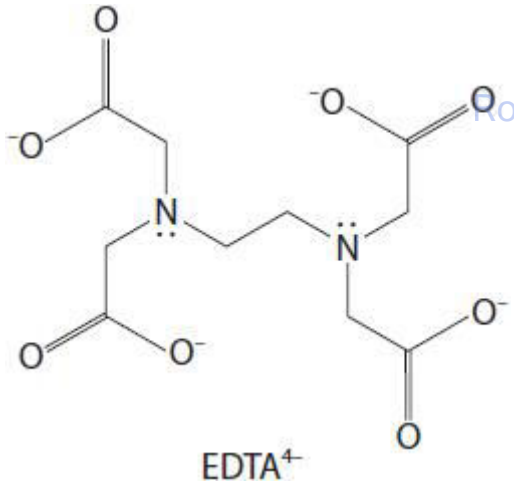
(c) We require small amounts of chromium(III) ions, Cr^{3+}, in our diet as they are important in the breakdown of fats, and many other body processes. However, too high a concentration of Cr^{3+} is toxic.

Blood should not contain more than 4 parts per billion by mass of Cr^{3+}.
The concentration of Cr^{3+} in a blood sample was 1.345×10^{-7} mol dm^{-3}.

(i) Show, by calculation, that the Cr^{3+} level in this blood sample exceeds the safe limit.
[1 billion = 1×10^{9} Assume the density of blood is 1.0 g cm^{-3}]

(2)

(ii) $EDTA^{4-}$ has been considered as a potential treatment for Cr^{3+} poisoning.



$EDTA^{4-}$

Suggest how $EDTA^{4-}$ would act to treat Cr^{3+} poisoning, giving a reason why $EDTA^{4-}$ would be preferred to other reagents such as 1,2-diaminoethane, $NH_2CH_2CH_2NH_2$.

(2)

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(Total for question = 17 marks)

Q101.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement about a catalytic converter used to reduce toxic emissions from internal combustion engines is **not** true?

- ☒ **A** the catalyst contains platinum
- ☒ **B** CO and NO are absorbed by the catalyst
- ☒ **C** after the reaction, desorption of CO_2 and N_2 takes place
- ☒ **D** it is a heterogeneous catalytic reaction

(Total for question = 1 mark)

Q102.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

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Which is **incorrect** for H_2O, OH^- and NH_3?

- ☒ **A** all can act as bidentate ligands
- ☒ **B** all can act as electron pair donors
- ☒ **C** all can form dative covalent bonds
- ☒ **D** all can take part in complex ion formation

(Total for question = 1 mark)

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Q103.

Cis-platin, $PtCl_2(NH_3)_2$, is used in cancer treatment as it inhibits DNA replication.

(a) State the shape of the *cis*-platin complex.

(1)

.....

(b) Draw the isomer *trans*-platin.

(1)

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(c) In human cells, the two chloride ions are replaced by water molecules in a ligand exchange reaction as shown.

$$PtCl_2(NH_3)_2 + 2H_2O \rightleftharpoons [Pt(NH_3)_2(H_2O)_2]^{2+} + 2Cl^-$$

Suggest why this occurs.

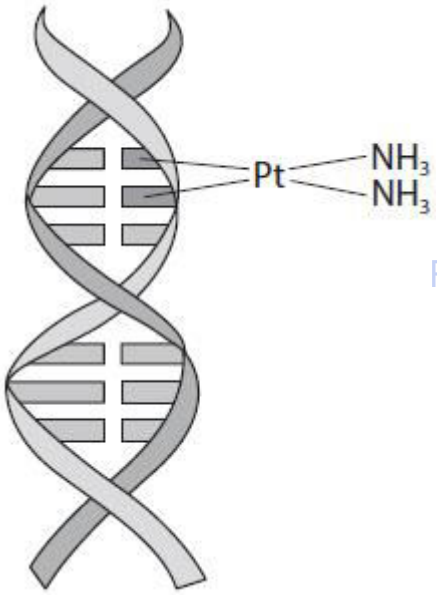
(2)

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(d) DNA forms two dative covalent bonds with *cis*-platin as shown.



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(i) DNA acts as a ligand in the formation of the complex shown in the diagram.
Name this type of ligand.

(1)

.....

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(ii) Explain, by considering the equation shown, why this process is thermodynamically feasible.

$$PtCl_2(NH_3)_2 + DNA \rightarrow [Pt(NH_3)_2(DNA)]^{2+} + 2Cl^-$$

(2)

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(Total for question = 7 marks)

Q104.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

What is the colour of an aqueous solution of V^{2+} ions?

- ☐ **A** blue
- ☐ **B** green
- ☐ **C** purple
- ☐ **D** yellow

(Total for question = 1 mark)

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Q105.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

What is the electronic configuration of a Zn^{2+} ion?

- ☐ **A** [Ar] $3d^{10}$
- ☐ **B** [Ar] $3d^{10}\ 4s^2$
- ☐ **C** [Ar] $3d^9\ 4s^1$
- ☐ **D** [Ar] $3d^8\ 4s^2$

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(Total for question = 1 mark)

Q106.

*A solution containing tetrachlorocuprate(II) ions, $[CuCl_4]^{2-}$, is treated with dilute aqueous ammonia.

The ammonia solution is added gradually.

The following observations are made:

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yellow solution → pale blue solution → pale blue precipitate → deep blue solution

Explain the changes occurring in this sequence.

Your explanation should refer to the formulae, shapes and colours of the species present at each stage.

Detailed explanations of how the colours of transition metal complexes occur are not required.

(6)

(Total for question = 6 marks)

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